

The Crucible: What Survived the Fire

Five Forms of Error and the Rebuilding of the Instrument

Volume 12: The New Universe of Eloquent Resonance

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KishLattice Geometric Harmonic Spectroscopy

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Preface: The Instrument Turned On Itself

Where Volume 11 Left Off

Volume 11 reported fifty-two active domains across forty-one orders of magnitude, twenty-four point nine million records, and a multi-attribute convergence on the Gaia stellar population that we described as the first of its kind in this programme. It closed with a theorem, two errata, and a phase-scramble gate built specifically to answer an objection we had raised against ourselves.

This volume reports what happened when we pointed the same scepticism at the machine that produced those numbers.

What Volume 12 Is

Volume 12 is not an expansion. It is an audit, and the audit found five distinct ways in which our own instrument could manufacture a result that looked like physics.

The count of domains reported as **STRONG** fell from twenty-eight to nine. One headline result — stellar transverse velocity locking at $15/\pi$ – $16/\pi$, carried in this programme since Volume 5 — is withdrawn. Two of four falsification controls turned out to be demonstrating something other than what they were built to demonstrate. A designated null was found to reproduce the very domain it was meant to null. The register assignment of every domain in the survey was shown to depend on a choice of physical unit that had never been pre-registered.

None of that was found by a critic. All of it was found by running measurements designed to test hypotheses that turned out to be wrong.

Why the Title

A crucible does not destroy what is placed in it. It separates. The slag and the metal go in together and come out apart, and the metal that emerges is worth more precisely because you watched the slag leave.



Figure 1: Twenty-eight went into the fire. Nine came out. This volume is an account of the fire. -Timothy John Kish

The Direction of Every Error

There is one structural fact about this audit that deserves to be stated before any of the detail, because it governs how the rest should be read.

Confirmed Signal

Every artifact form identified in this volume biases the measurement *against* the framework. Form C displaces register assignment. Form D depresses every negative verdict. Form E inflates the significance of quantised domains without creating structure where none exists. The residual left by our own correction is negative-signed at every register we have measured.

None of the five forms is capable of manufacturing a positive peak from nothing. Every one of the twenty-eight STRONG domains in Volume 11 was scored into a headwind.

That asymmetry is not a comfort we have arranged for ourselves. It is a consequence of the mechanism of each defect, demonstrated in Chapter 4, and it is the reason the nine surviving domains mean more after this audit than the twenty-eight did before it.

The Team, and What the Seats Are For

The Aurora Protocol assigns distinct roles that cannot be collapsed. Atlas designs experiments and cannot rule. Mondy rules and cannot design. Lyra builds lakes and reads results with an enthusiasm that requires balancing. Vera reports. Phoenix documents. Timothy decides.

That structure earned its keep in this volume in a way worth recording. Of the findings reported here:

- Two were rediscoveries of material already published in Volume 11's errata, filed by Atlas and caught by Mondy. A standing rule now requires the errata register to be searched, and the result cited, before any engine finding is filed.
- One was a validation reported as passing against a test fixture whose documentation claimed one thing and whose contents were another. Atlas found it; a standing rule now requires every fixture to assert its own tuning parameter at runtime.
- One was a mechanism proposed by the referee, measured by the architect, and found to be wrong — which produced a better mechanism and a larger finding than the original hypothesis would have.
- One was a bound the referee had established correctly for one object and the architect extended incorrectly to another. The referee's disproof used the architect's own table.

Three wrong hypotheses produced three real findings. That is approximately the ratio a working instrument should show, and it is the strongest argument we can make that the process is not merely decorative.

What This Volume Asks of the Reader

It asks patience with Chapter 4, which is the longest and the most technical, because the five artifact forms are the substance of the volume and each requires its mechanism to be understood rather than asserted.

It asks that Chapter 8 be read as carefully as Chapter 6. Chapter 6 reports what survived. Chapter 8 reports what the survivors do *not* establish, and the second is the more important of the two.

And it asks that the errata in Chapter 7 be read as the record of a method working, not as a list of failures. A programme that publishes only its confirmations has told you nothing about its error rate.

— Timothy John Kish, August 2026

Chapter 1

Background: Where Error Lives

1.1 Two Ways to Be Wrong

Volume 11 opened by contrasting the old world of emission spectroscopy with the new world of geometric harmonics. Volume 12 opens with a contrast of a different kind, and it is not between us and anyone else. It is between two ways that a measurement can go wrong.

[Old World:] *In a controlled experiment, the apparatus is built to isolate the phenomenon. The risk is that the apparatus itself produces the signal. The experimenter owes the reader an argument that the control did not manufacture the result.*

[New World:] In an archival survey, the data arrives as it is. There is no apparatus to bias. The risk moves downstream: it lives in *promotion* — in how a public catalogue becomes a scalar array. The surveyor owes the reader an argument that the pipeline did not manufacture the result.

Neither is cleaner. They are exposed in different places, and the honest question is not which method is purer but whether the practitioner went looking in the place where their own method hides its errors.

1.1.1 A Worked Example From the Other Side

In 2026 a team at South China Normal University performed the first direct experimental test of Feynman’s two 1948 path-integral postulates [18], measuring single-photon probability amplitudes across more than 1.4 million discrete optical paths — $17^5 = 1,419,857$ — and reconstructing the path probability amplitudes in full [19, 20]. The work reports what it describes as unprecedented fidelity in propagator measurement, achieved by suppressing the environmental phase noise that had limited the same group’s earlier propagator experiment [21].

A hostile reading is available: they suppressed noise until the answer appeared. We want to be explicit that this reading is wrong, and that understanding *why* it is wrong is instructive for our own position.

Suppressing decoherence is not tuning toward a preferred outcome. Decoherence is a known physical confound with an independent theory of its own, and removing it is ordinary good practice — the same category of move as shielding a detector from cosmic rays. The distinguishing question is whether the suppression was specified in terms of the confound or in terms of the desired answer. Theirs was the former.

Note also what that team did with its own bar: they identified the systematic that limited their earlier result, suppressed it, and reported the improvement against their own prior benchmark

rather than against a threshold chosen afterward. That is the same move this volume makes in Chapter 3 when the STRONG threshold is restored from 3.0 to 5.0.

1.1.2 Our Exposure, Stated Plainly

Ingesting an entire public catalogue prevents *selection bias*. One cannot cherry-pick favourable objects if every object is taken. This has been a genuine strength of the programme and it remains one.

It does not prevent *promotion artifacts*, and promotion is precisely where ours were.

- A tidal domain whose reported lock rests on 3,515 records carrying the value 12.0000 hours, when the physical semidiurnal period is 12.4206 hours and does not lock at that register.
- A transit-timing domain whose register is determined by the fact that the source reports whole minutes.
- A stellar kinematic lake in which 27.5 percent of records carry transverse velocities below 0.19 km/s, where the expected fraction for the underlying population is approximately 0.002 percent.

None of those is a selection effect. Every one of them entered before we saw the data.

1.2 Adopting the Look-Elsewhere Correction

The framework scans 23 registers per domain. Across the enabled survey this is approximately fifty domains, giving of order

$$T = 23 \times 50 = 1,150 \text{ trials.}$$

A local significance must therefore be converted to a global one:

$$p_{\text{global}} = 1 - (1 - p_{\text{local}})^T.$$

The consequence is not cosmetic. At $T = 1,150$, a local $z = 5.0$ corresponds to a global significance of approximately 3.41σ . To claim a global 5σ across the full trial space requires

$$z_{\text{local}} \geq 6.22. \tag{1.1}$$

Methodological Note

This correction was absent from Volumes 5 through 11. Its absence was a genuine gap, and had a referee found it before we did, our STRONG threshold would have appeared to be set where the results happened to be.

Volume 12 adopts $z_{\text{local}} \geq 6.22$ as an entry gate applied to *every* domain in the survey, not reserved for a headline claim. The threshold is derived from the trial count rather than chosen; a reader can recompute it in one line.

We note also that our z is not a Gaussian test statistic in the sense a particle physicist means. It is a lock rate compared against a Monte Carlo null of finite trial count, and it carries a run-to-run

uncertainty of its own, quantified in Chapter 3. We therefore report “exceeds a look-elsewhere-corrected local z of 6.22” rather than “ 5σ ”, because the former states what was measured and invites checking.

1.3 The Eleven-Volume Arc, and Where It Now Stands

Volumes 1 through 11 established the geometric modulus $k_{\text{geo}} = 16/\pi = 5.0929581789$ as a scalar coordinate, built the lake architecture, developed the three-tier null protocol, and closed Form A by theorem. Volume 12 does not withdraw that arc. It corrects the instrument that measured against it, and in doing so it re-states what the arc has established and what it has not.

Stands	Restated or withdrawn in this volume
The scalarization and the modulus	The unit in which each quantity is expressed, never pre-registered
Form A closed in 1D by theorem	Form A does not extend to 2D; its real-anchor sweep requires re-running under the corrected statistic
The lake architecture and chain of custody	Promotion standards, now requiring native precision and declared plausibility ranges
quantum_transitional at $16/\pi$	stellar_kinematic at $15/\pi$ – $16/\pi$, withdrawn
Honest nulls as first-class results	Every negative verdict in Volumes 5–11, now known to be depressed by Form D

Table 1.1: What the arc establishes after the audit.

Chapter 2

The Regression That Started It

2.1 A Routine Gate

The audit began as a formality.

Before any two-dimensional work could be trusted, the referee required that the rebuilt Volume 12 engine reproduce two known results in one dimension: the NIST atomic transition lake (`L_emission_nist`, domain `quantum_transitional`) and the Gaia transverse velocity lake (`s2_stellar_kinematics`, domain `stellar_kinematic`). Two lakes, six minutes of compute, a box to tick.

The run of 16 August 2026 returned what looked like a clean pass. `quantum_transitional` locked at $16/\pi$ with a chaos z of +28.5. `stellar_kinematic` returned $15/\pi$ at +93.1 and $16/\pi$ at +92.1, consistent with the band this programme had reported across six volumes.

2.2 One Number in Two Places

The domain summary reported a mean scalar for `stellar_kinematic` of 1.0930.

The lake’s own metadata, written at promotion in April 2026, contained the field `meta.global_mean` with the value 1.092971.

Self-Correction

The pipeline’s computed domain mean matched a constant already stored inside the lake, to six decimal places. A computed statistic should not reproduce a stored one unless the pipeline is reading the stored one.

Reconstructing the arithmetic from a single record settled it. The lake stores three values per star:

$$\text{val_raw_kms} = 54.72644642484259 \quad (\text{transverse velocity, km/s}) \quad (2.1)$$

$$\text{scalar_raw} = 2.469781 \quad (2.2)$$

$$\text{scalar_klc} = 2.670401629977723 \quad (2.3)$$

The middle value is exactly the scalarization of the first:

$$\frac{\ln(1 + 54.72644642484259)}{\ln(16/\pi)} = 2.469781.$$

The third is the second with a per-sector offset applied:

$$\text{scalar_klc} = \text{scalar_raw} - \text{sector_mean} + \text{global_mean} = 2.469781 - 0.892351 + 1.092971 = 2.670402.$$

A per-sector additive shift is mean-preserving by construction, so the mean of the shifted scalars equals `global_mean` exactly. The pipeline had consumed `scalar_klc` — a *sector-normalised, precomputed* scalar — while the configuration file specified a field named `v_perp` that does not exist in that lake at any nesting depth.

2.3 The Silent Substitution

The mechanism was a fallback layer in `scalarize.py`. When the named handler and the configuration dispatch both returned zero, the engine adopted the lake’s stored `scalar_kls` without comment. The domain then reported `nonzero = n` and every downstream diagnostic showed a healthy read.

Methodological Note

This is the most dangerous class of failure available to a pipeline: not a wrong number that announces itself as wrong, but a wrong number that produces a perfectly healthy diagnostic trail. A domain reading a quantity other than the one it is configured to read is worse than a domain reading zeros, because zeros are visible.

Under the pre-patch engine, six domain groups covering **4,083,300 records** were found to be on this fallback. The condition producing it — a configured field absent from the lake, and no named handler claiming that domain — has been constant since Volume 5.

2.4 The Question the Re-Run Was Built to Answer

The engine was patched (Chapter 3) so that the configured field is resolved through nested payloads and the fallback becomes a hard failure. One question was pre-registered before the re-run:

Formal Pre-Registration

Does the $15/\pi$ – $16/\pi$ stellar band survive being computed from raw transverse velocity rather than from the sector-normalised precomputed scalar?

Expected: `quantum_transitional` reproduces bit-identically, since it resolves its field at top level and never touched the fallback. `stellar_kinematic` should shift its domain mean off 1.092971, and its negative minimum of -0.1935 should disappear, that minimum being an artifact of the sector offset rather than a negative velocity.

Both predictions held. `quantum_transitional` returned $n = 189,330$, mean 20.6754, range $[9.4501, 23.6047]$ — identical to the prior run. `stellar_kinematic`’s range moved from $[-0.1935, 4.4446]$ to $[0.0000, 4.2441]$.

And the band did not survive. Computed from raw velocity, the domain returned negative z at all twenty-three registers, the largest being -70.7 .

That result, and the two weeks of work it triggered, is the subject of this volume.

Chapter 3

The Engine, Rebuilt

3.1 What Changed, and What Did Not

Five changes were made to the engine. It is important to state at the outset what was *not* changed: the mathematics of the scalarization is untouched. All four formula branches — `log_standard`, `log_inverse`, `log_ratio`, `log_volumetric` — are algebraically identical to Volume 11.1 and were unit-tested as such before adoption.

Every patch was applied by exact-match assertion against the file whose MD5 matches the published Volume 11 engine fingerprint. No source line was retyped.

3.1.1 Change 1: Field Resolution Through Nested Payloads

Volume 11.1 located a configured field by checking the record top level, then `_raw_payload`, then `payload`, using an `or-chain`. Promoted lakes nest their raw values at varying depth — the T-series sits four levels down after repeated re-promotion — so configured fields were frequently unreachable.

The `or-chain` carried a second defect: a stored value of 0.0 is falsy in Python and fell through as though the field were absent. All checks are now `is None`.

3.1.2 Change 2: The Fallback Made Loud

Every record’s scalar is now attributed to the route that produced it. The legacy precomputed-scalar substitution is counted, named, and under `KLGHs_STRICT=1` aborts the run with the offending domains listed.

```
LEGACY FALLBACK FIRED.
These domains did NOT compute from their configured field.
    biology          cosmology          orbital
    planetary        stellar          stellar_rotation
ABORTING: legacy fallback fired under KLGHS_STRICT=1.
```

Listing 3.1: Dispatch attribution, run of 23 August 2026

All six traced to a single cause: the scalarizer dispatches on the domain field stored *inside each record*, and several promoted lakes carry stale or pooled domain strings. Record counts confirmed it exactly — `stellar` = 3,979,697 = `s1` (2,000,000) + `s4` (1,979,697); `stellar_rotation` = 67,311 = `k1` (64,784) + `k2` (2,527); and so on for all six.

3.1.3 Change 3: Common-Support Restriction

This is the substantive statistical change and Chapter 4 develops its motivation. The lock test computes, for scalar s at register N ,

$$r_p = \frac{s}{N/\pi} \times C, \quad C = 24, \quad (3.1)$$

and counts a record as locked when $|r_p - \text{nearest}| < 0.05$, where $\text{nearest} = \max(1, \text{round}(r_p))$.

The clamp to 1 creates a region of the scalar axis that is *mechanically non-lockable*. Its boundary is an algebraic identity:

$$s_{\text{floor}}(N) = \frac{1}{2} \cdot \frac{N/\pi}{C} = \frac{N/\pi}{48} = \frac{1}{2} (\text{modular period}). \quad (3.2)$$

Exactly half a period, at every register, for every domain. The correction excludes all records with $r_p < 0.5$ from *both* the real data and the null, at every register, and publishes the excluded fraction. The cut sits at the anti-node, maximally distant from any lock window, so it neither adds nor removes locked records preferentially.

3.1.4 Change 4: The Offset-Sweep Baseline

Common support removes the clamp discontinuity but not the residual shape mismatch above the floor. The residual is measured rather than modelled, using machinery the programme already possessed:

$$z_{\text{signal}}(N) = z(N) - \text{median}_{\phi} z(N, \phi), \quad (3.3)$$

where ϕ sweeps the grid phase across one full period. The reported discriminant becomes **peak minus median** rather than peak versus zero. Because the uniform null's lock rate is invariant under a shift of grid phase, the null is computed once per register and only the real rate is swept; the sweep is nearly free.

3.1.5 Change 5: Numerical Correctness

The engine computed $\ln(1.0 + x)$ directly. In IEEE double precision, $1.0 + x$ evaluates to exactly 1.0 for $x \leq 1.11 \times 10^{-16}$, so any such value produced a scalar of exactly zero. The volumetric branch is worse, forming $\ln(1.0 + x^3)$, which absorbs at $x \leq 4.81 \times 10^{-6}$ — a threshold reached by `kinematic_wrongbox`, whose input velocities have a measured minimum of 4.88×10^{-13} km/s.

All three sites now use `log1p`, which is exact in that regime and identical elsewhere.

3.2 The Run-to-Run Uncertainty of z

The pinch stage compares a deterministic real lock rate r against a null estimated from T random trials with sample mean m and standard deviation s :

$$z = \frac{r - m}{s}.$$

Since r does not vary between runs, all run-to-run variation enters through the finite-trial null estimate. Propagating both terms:

$$\text{sd}(z) \approx \sqrt{\frac{1}{T} + \frac{z^2}{2T}} \approx \frac{z}{\sqrt{2T}}. \quad (3.4)$$

Monte Carlo confirms it: at $T = 100$ and $z \approx 92$, simulation gives $\text{sd}(z) = 6.77$ against a predicted 6.51; at $T = 1000$, 2.07 against 2.06.

Methodological Note

At the standing value $T = 100$, a reported z carries an intrinsic run-to-run uncertainty of roughly 7 percent of its own magnitude. Two runs at an identical engine fingerprint on an identical master will disagree.

Single-run z is withdrawn as a citable quantity from Volume 12 forward. Register occupancy across repeated passes, mean $\pm \sigma$ per register, and the uncertainty band are reported together.

An immediate consequence: the Volume 11 report of `stellar_kinematic` at $15/\pi$ (+98.3) versus $16/\pi$ (+94.0) separated two registers by 4.3 against an uncertainty of ± 6.8 . Those registers were never distinguishable, and naming either was reporting noise.

3.3 The Threshold

The Volume 11.1 code labelled a domain `STRONG` at $z > 3.0$, while the programme's pre-registered standard is $z \geq 5.0$. Two domains in the run of 23 August were printed `STRONG` without meeting it: `null_cosmology_null` at +4.1 — a designated null — and `biology_chirality` at +3.3.

The threshold is restored to 5.0, printed at run start, and superseded for survey-wide claims by the look-elsewhere value of 6.22 derived in Chapter 1.

3.4 Two-Sided Validation Before Adoption

Referee Ruling

A fix that has not been shown to fix the known artifact is not permitted to certify anything.
A fix that suppresses the artifact by suppressing sensitivity is not a fix.

The corrected statistic was validated against synthetic data before it was permitted to touch a lake. Four tests, expectations fixed in advance.

Test	Data	Uncorrected	Corrected
T1	right-skewed, no structure, 27.5% below floor, $n = 1.8\text{M}$	median -132.5	median -4.92
T2	same data, lock planted at $19/\pi$	masked	+182, correct register
T3	uniform, no structure	median $+0.72$	median $+0.58$
T4	uniform, planted lock	detected	detected

Table 3.1: Two-sided validation. T2 is the half that matters: the correction must not remove sensitivity along with the artifact.

With the offset-sweep baseline applied, the same structureless data returns a median z_{signal} of -1.62 with a maximum absolute value of 3.52 across all twenty-three registers — below the STRONG threshold everywhere. No structureless domain can reach STRONG under this statistic.

Self-Correction

An earlier report of this validation was made against a fixture whose documentation claimed 27 percent below-floor mass while its contents carried 10.37 percent, returning a median of -11.17 rather than the -132.5 characteristic of the real artifact. The fix had been validated against a milder disease than the one present.

Standing rule adopted: every validation fixture asserts its own tuning parameter at runtime and fails loudly on mismatch. Documentation is not permitted to stand in for a measurement anywhere in the chain.

Chapter 4

Five Forms of Error

4.1 What a Form Is

A *form* in this programme is a named mechanism by which the instrument can produce a result that looks like physics and is not. Forms are not mistakes; a mistake is fixed once. A form is a permanent property of the measurement architecture that must be screened for on every run, forever.

Volume 11 carried two. Volume 12 carries five.

	Form	Mechanism	Status
A	Transform artifact	The scalarization manufactures apparent structure from distribution shape alone	Closed by theorem, 1D only
B	Mundane concentration	Real clustering arising from scale coincidence or selection, not lattice geometry	Open objection
C	Preprocessing concentration	Structure introduced by an operation applied to the lake <i>before</i> the pipeline saw it	Erratum 4
D	Null-support mismatch	The null's lockable support does not match the data's	Erratum 4
E	Quantised support	Few distinct values; the lock reduces to whether those values land on nodes	Erratum 4

Table 4.1: The artifact taxonomy after Volume 12.

4.2 The Grid, and Why It Has a Floor

Every form below acts on the same object, so it is worth drawing it once.

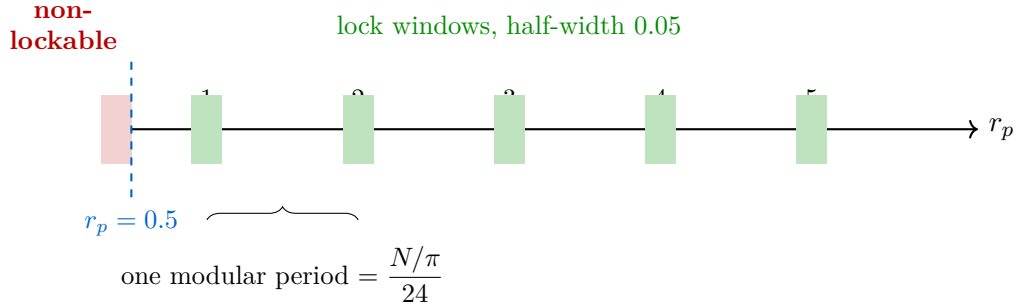


Figure 4.1: The modular grid. A record locks when its position r_p falls within 0.05 of an integer node. Because `nearest` is clamped to a minimum of 1, every record with $r_p < 0.5$ is compared against node 1 and can never satisfy the tolerance. That boundary sits at exactly half a modular period, at every register — Equation 3.2.

4.3 Form C — Preprocessing-Induced Concentration

Artifact Form

Form C. Structure introduced by an operation applied to the lake before the pipeline saw it. Invisible to all three nulls, because the nulls are generated after the operation and do not replicate it.

The Gaia kinematic lake stores a sector-normalised scalar, formed by subtracting a per-sector mean and adding the global mean. Three lakes in the survey carry `sector_normalized: true`: `s1_gaia_parallax`, `s2_stellar_kinematics`, and `g1_galaxy_kinematics`.

The below-floor mass of such a lake forms a coherent *block* occupying exactly half a modular period (Equation 3.2). A per-sector shift does not disperse that block — it translates it, and where it lands relative to the node is determined entirely by $\Delta_j \bmod \text{period}$.

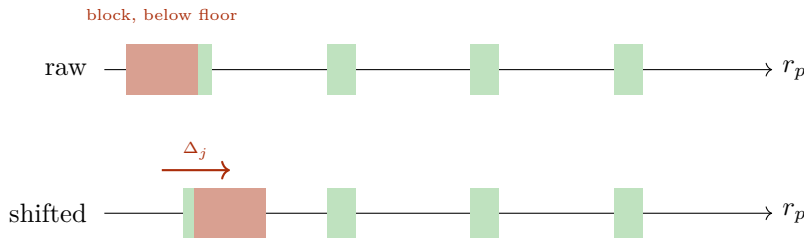


Figure 4.2: Form C. In raw form the half-period block sits below the floor and contributes no locks. A per-sector additive shift translates the whole block; if it lands on a node, a substantial fraction of the domain locks at once. Different sectors carry different offsets and light different registers.

Methodological Note

The decisive objection is not the mechanism but the null. The chaos null is generated by uniform sampling over the empirical range of the *normalised* scalars. It does not replicate the per-sector shift. The comparison being made was normalised-real against unnormalised-uniform: the null was blind to the single operation most capable of manufacturing the effect.

This is disqualifying independently of whether any artifact mechanism is demonstrated. A normalised scalar tested against an unnormalised null cannot support a lock claim.

Sector normalisation is defensible astrophysics — Gaia has genuine sky-position-dependent selection effects. The ruling is not that the operation is illegitimate. It is that if retained, it must be re-pre-registered as its own domain, with a null that applies the identical per-sector procedure to surrogate data.

4.4 Form D — Null-Support Mismatch

Artifact Form

Form D. The null’s lockable support does not match the data’s. A uniform null over the empirical range carries a small below-floor fraction; a right-skewed real distribution carries a large one. The null is structurally more lockable than the data it scores, and the resulting z measures that difference.

This is the form with the widest reach. It biased every negative verdict in eleven volumes.

4.4.1 The Measurement

Direct measurement of `s2_stellar_kinematics` against the floor:

	below the $16/\pi$ floor
Real distribution (1,808,145 records)	27.53%
Uniform chaos null over the same range	2.50%

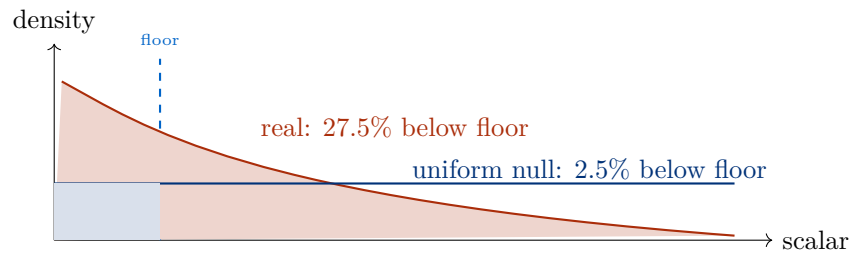


Figure 4.3: Form D. Both arms span the same range, so the null appears fair. They place very different amounts of mass in the mechanically non-lockable region, and the resulting z is a measurement of that difference rather than of harmonic structure.

4.4.2 The Demonstration

If the entire signal is floor mismatch, then predicting the lock rate from the below-floor fraction alone — assuming *no harmonic structure whatsoever* — should reproduce the observed profile:

$$\widehat{\text{lock}}(N) = (1 - f_{\text{below}}(N)) \times 0.10. \quad (4.1)$$

Confirmed Signal

Against the 23-register profile of `stellar_kinematic`:

mean absolute error	1.23%
maximum error	2.67%
$\text{corr}(f_{\text{below}}(N), \text{observed lock rate})$	−0.99708

Every register. No structure assumed. The monotonic decline in observed lock rate from 0.0828 at $4/\pi$ to 0.0669 at $26/\pi$ tracks the monotonic rise in below-floor fraction from 16.7% to 32.5%, because a higher register has a higher floor.

4.4.3 The Sign Bound

Form D has a property that limits the damage, and it is worth proving rather than asserting. The below-floor fraction is the domain’s own cumulative distribution function evaluated at the floor:

$$f_{\text{below}}(N) = F\left(\frac{N}{48\pi}\right). \quad (4.2)$$

A CDF is non-decreasing, and $N/48\pi$ is increasing in N . Therefore f_{below} is monotonically increasing in N for *any* distribution.

Confirmed Signal

A monotonic function has no local maxima. **Form D can produce a smooth negative baseline deepening with N , but it cannot manufacture a peak at any register.**

Consequently every positive claim in the survey was scored against a headwind, and every *negative* verdict — “noise floor”, “anti-signal” — is in question.

4.4.4 The Residual

Common support removes the clamp discontinuity. It does not remove the shape mismatch above the floor: the real distribution remains skewed while the null remains uniform. On severity-matched structureless data the residual after correction reaches $|z| \approx 10.6$, and — unlike Form D proper — it is *not* monotonic. It is an alignment-against-grid quantity, which is the same kind of object a lock is, and can therefore carry local structure.

That is why the offset-sweep baseline of Chapter 3 is measured per register rather than per domain. With it applied, the residual falls to a maximum of 3.52.

4.5 Form E — Quantised Support

Artifact Form

Form E. The domain takes a small number of distinct values. The lock statistic reduces to whether those specific values land near nodes, and the record count inflates z far beyond the information content.

4.5.1 The Tidal Domains

`planetary_atlantic` reports a lock rate of 0.5054 at $17/\pi$ — more than half the lake. The domain carries **fifteen distinct scalar values**.

Domain	n	distinct	locked mass on top values
<code>planetary_atlantic</code>	6,973	15	99.7% on 12.0000 h
<code>planetary_gulf</code>	6,927	23	96.2% on {12.0000, 8.0000}
<code>planetary_pacific</code>	6,838	22	90.6% on 14.0000 h
<code>planetary_indian</code>	4,929	40	86.2% on 14.0000 h ± 1 s

4.5.2 The Decisive Calculation

The principal lunar semidiurnal constituent M_2 has a period of **12.4206 hours**, not 12. A lake reporting integer hours is reporting a rounded quantity. Testing both:

Value	scalar	r_p at $17/\pi$	distance to node	
12.0000 h (what the lake contains)	1.575658	6.9883	0.0117	locks
12.4206 h (true M_2)	1.595130	7.0751	0.0751	does not lock

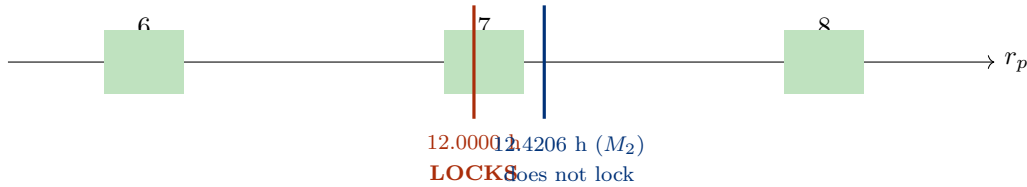


Figure 4.4: Form E made concrete. Two tick marks 0.42 hours apart at $17/\pi$. The rounded integer falls inside the lock window; the physical semidiurnal period falls outside it. A STRONG classification on 3,515 records hangs on which of the two the lake recorded.

4.5.3 The Arithmetic of the Modulus Itself

`orbital_ttv` — transit timing variation, $n = 290,458$, reported STRONG at $16/\pi$ — carries its top five locked values at raw 1, 3, 7, 15, 31 minutes. Those are $2^m - 1$, and their scalars are exact integer multiples of $\ln 2 / \ln k_{\text{geo}} = 0.425803$.

For such values,

$$r_p = m \cdot \frac{24\pi \ln 2}{N \ln k_{\text{geo}}} = m \cdot \frac{32.1048}{N}, \quad (4.3)$$

and 32.1048 lies 0.32 percent from $32 = 2^5$. The ratio is therefore near-integer at $N = 4, 8, 16$ — the powers of two dividing 32 — and nowhere else in the family.

N	$32.1048/N$	multiples of $2^m - 1$ that lock	observed Δ
16	2.0065	$m = 1 \dots 7$ (seven)	+0.0473
8	4.0131	$m = 1, 2, 3$ (three)	+0.0351
4	8.0262	$m = 1$ (one)	-0.0194

Table 4.2: The per-multiple error accumulates with m , so the higher register retains more multiples. The observed ordering is exactly the count of surviving multiples.

Modelling `orbital_ttv` as *nothing but integer minutes* with a $1/x$ weight, and assuming no harmonic structure at all, reproduces the observed profile at a correlation of +0.804, with the same peak register and four of the top five.

Methodological Note

This is a property of the modulus, not of the data. Any binary-commensurate quantisation, in any domain, will favour registers 4, 8 and 16. The framework’s principal register is the one most vulnerable to this artifact.

It does not make $16/\pi$ results wrong. It means the evidential burden on a $16/\pi$ claim is higher than on any other register, and that continuous-support verification is mandatory before any $16/\pi$ classification. A continuous domain has no quantisation grid and cannot be reached by this mechanism. `quantum_transitional` carries more than 50,000 distinct values and is unreachable.

The screen generalises beyond base two: a grid of spacing g produces preferred registers wherever $24\pi \ln g / (N \ln k_{\text{geo}})$ is near-integer. For decimal rounding, $g = 10$ gives a constant of 106.65, favouring registers near 4 and 5. Round-number decimal quantisation is far more common in promoted data than binary.

4.6 The Screens Now Standing

Screen	Catches
Dispatch attribution	Form C at source: a domain reading a precomputed scalar rather than its configured field
Common support + baseline	Form D and its residual
Kept-fraction per register	Under-resolved domains; out-of-range reporting
Periods-spanned	Domains with too few nodes across their entire range
Distinct-value / grid screen	Form E, including grids other than base two
Node-occupancy histogram	Form C and E: locked mass concentrated on one or few node indices
Look-elsewhere correction	Trials-factor inflation across 1,150 register-domain tests

Table 4.3: Every screen is run on every domain on every run, and its result published.

Chapter 5

The Crucible: Lake by Lake

5.1 What the Fire Showed

The forms of Chapter 4 are mechanisms. This chapter is what they did to specific lakes. Each entry states what was found, how it was found, and what the lake now requires.

5.1.1 `s2_stellar_kinematics` — 27.5% Non-Measurements

The Gaia transverse velocity lake carries 1,808,145 records. Measured against the $16/\pi$ floor of 0.188535 km/s:

- **497,831 records (27.53%)** sit at or below it.
- 125,726 records (6.95%) carry $v_T < 0.005$ km/s.
- The minimum is 4.88×10^{-13} km/s.

For a stellar population with velocity dispersion $\sigma \approx 30$ km/s, the expected fraction below 0.19 km/s is approximately 2×10^{-5} — about 0.002 percent. The lake contains 27.53 percent. That is four orders of magnitude, not a tail effect.

Confirmed Null

A transverse velocity of 10^{-13} km/s is not a measurement. These are almost certainly proper motions at or beneath Gaia’s detection floor, promoted without filtering. The identical minimum appears in `s2_wrongbox`, confirming a shared source.

Required: a physical validity criterion, pre-registered — proper motion above a fixed multiple of its own quoted Gaia uncertainty. Not a threshold chosen at the scalar level after its effect is known. If `pmra_error` and `pmdec_error` are absent from the promoted lake, that is a promotion defect and the lake rebuilds from source.

Note: the filter does not fix Form D. Removing the visibly absurd 6.95% still leaves roughly 20% below floor. These are two separate defects that happen to overlap in this lake.

5.1.2 The Tidal Lakes — Rounded to Integer Hours

Covered in Chapter 4. **Required:** rebuild from NOAA CO-OPS timestamps at native precision. The precision exists upstream — `t2_planetary`’s raw payload records times to the minute (04:59

to 09:40, giving 4.683333 hours). If the lock survives at native precision it is real; if it vanishes, Form E is confirmed. Suspension, not withdrawal, pending that run.

5.1.3 `orbital_ttv` — Whole Minutes

Same remedy, same falsification structure. Suspended pending a native-precision rebuild.

We record one honest residual: the quantisation model under-predicts the observed peak at $17/\pi$, where the locking multiples are $m = 9$ and $m = 18$, both low-weight. Something at $17/\pi$ is not explained by quantisation. That is precisely what a native-precision re-run would resolve, and it is stated here rather than smoothed over by the correlation coefficient.

5.1.4 `stellar_spindown` — Eleven Orders of Magnitude Annihilated

Pulsar spin-down rates span 1.17×10^{-21} to 5.49×10^{-10} s/s. Because $\ln(1+x) \approx x$ for $x \ll 1$, and $x_0 = 1.0$, the entire domain maps to scalars of order 10^{-10} — nine orders of magnitude beneath the lock floor of 0.106.

The domain carries **zero information**. It cannot have been an honest null in any volume in which it appeared, because there was nothing to null.

This is the same scale annihilation that disqualified two wrongboxes (below), occurring in a domain treated as real data. Its root is broader than one lake: $x_0 = 1.0$ **is set for all domains**, regardless of the natural scale of the quantity. The parameter exists to fix the reference scale and has never been used.

5.1.5 `n4_cosmological` — A Null That Reproduces Its Own Domain

`t4_cosmological` carries `d_proxy`, a Hubble distance cz/H_0 in Mpc, over a redshift range of $z \in [0.01000, 0.01596]$ — implying a scalar range of $[2.32224, 2.60416]$.

`n4_cosmological`, the *designated null* for that domain, carries a measured scalar range of $[2.32227, 2.60422]$.

Erratum

Agreement to 3×10^{-5} and 6×10^{-5} . The null for the cosmology domain has the same scalar distribution as the cosmology data.

Both lock at $15/\pi$: `cosmology` at +18.7 and `null_cosmology_null` at +11.3. The null locked because the null *is* the signal. Every statement of the form “the null did not lock” that rests on this lake is void.

Required: rebuild or withdraw before either domain enters any test.

5.1.6 `t4_cosmological` — Under-Resolved

The same measurement exposed a second problem. That scalar range spans 0.282 units. At $15/\pi$ the modular period is 0.19894, so the entire domain spans **1.42 modular periods**. It locks at $15/\pi$ with roughly one and a half nodes available across its whole range.

That places it with `orbital_eccentricity` (1.93 periods) and `galactic_sersic` (2.01 periods). A register identification on two nodes is not a distributional finding.

5.1.7 chemistry — Not a Broken Lake

Reported early in the audit as requiring a rebuild, on the evidence that `scalar_klc` and `_raw_payload.scalar_invariant` were bit-identical, implying no log transform was applied.

The lake is intact. The named handler `scalarize_chemistry` returns the stored invariant verbatim, by construction. The passthrough is in the engine, not the data. Recorded here because a rebuild was nearly ordered on a misdiagnosis.

5.2 The Wrongbox Tier

A wrongbox applies deliberately *wrong* mathematics to *right* data, and is expected not to lock. The tier was believed to comprise four working controls.

Wrongbox	below-floor mass	Verdict
<code>stellar_wrongbox</code>	87.7% $\rightarrow$ 99.6%	scale annihilation, not projection wrongness
<code>kinematic_wrongbox</code>	$\approx$ 38–42%	same
<code>orbital_wrongbox</code>	$\approx$ 3%	genuine control
<code>materials_wrongbox</code>	$\approx$ 1%	genuine control

Referee Ruling

Rule 1.5, second clause. A wrongbox is a valid control only when its wrong projection does not shrink the scalar below the lock floor. `log_inverse` of a large quantity and `log_volumetric` of a sub-unit quantity are *scale annihilators*, not wrong projections. The retained fraction under common support must be published for every wrongbox, and any wrongbox retaining less than a pre-registered fraction is reported OUT-OF-RANGE, never COLLAPSE.

The direct evidence is stark. In the canonical run, `stellar_spindown` $\times$ `stellar_wrongbox` produced the **third-highest cross-domain pinch in the entire table**. `stellar_spindown` carries no information; `stellar_wrongbox` is 99.6% below floor. Two informationless domains, agreeing at rank three.

`orbital_wrongbox` is the template: cubed orbital periods spanning six orders of magnitude, never approaching the floor, returning real ≈ 0.098 against chaos ≈ 0.098 , $\Delta \approx 0$. That is a wrongbox doing its job, and it is the only clean demonstration in the tier.

5.3 The New Lake Boundaries

The following become standing promotion requirements. No lake is admitted without them.

1. **Native precision.** A lake states the precision of its source and promotes at it. Rounding is a promotion decision and must be declared, not inherited.
2. **Declared physical plausibility range.** Every lake states the expected range of its quantity from independent physics, and the fraction of records outside it, before admission. The distributional litmus test cannot see physical implausibility: a lake roughly a quarter composed of non-measurements passed it.

3. **Measured object overlap.** Where two lakes may describe the same objects, the overlap is *measured*, not assumed. Where partial, the intersecting records are removed from one pre-registered side rather than a domain being discarded.
4. **Declared x_0 .** The reference scale is drawn from independent physics, cited, and filed before the run — never adjusted after a register is seen.
5. **Periods spanned, published.** Any domain below the pre-registered minimum is reported as insufficient-resolution rather than given a register.
6. **Distinct-value count and grid detection, published.**

5.3.1 A Worked Example of Requirement 3

`t4_cosmological` (SDSS DR16 redshift) and `g1_galaxy_kinematics` (SDSS DR16 velocity dispersions) were candidates to be collapsed into a single family on the suspicion of shared objects. Rather than assume, the overlap was matched on sky position at 1-arcsecond tolerance:

matched pairs	971
fraction of <code>t4_cosmological</code>	19.42%
fraction of <code>g1_galaxy_kinematics</code>	0.053%
redshift range overlap	100% of A, 0.6% of B

Partial, not fatal. The 971 intersecting records are removed from `t4_cosmological` and both domains are retained as independent families. Collapsing an entire domain on a suspicion would have discarded a lake over a fifth of a small sample.

We note the caveat the measurement also surfaced: zero object overlap would not have implied independence. Two disjoint samples drawn from the same flux-limited survey over the same redshift range are correlated through selection.

Chapter 6

The Controlled Same-Object Experiment

6.1 A Liability Turned Into a Prediction

Volume 11 reported multi-attribute measurement of a single population as a strength. Volume 12 must correct that: measuring one population several ways produces results correlated *by construction*. Stellar colour against stellar luminosity is the Hertzsprung–Russell relation, known since 1911. Galaxy dispersion against galaxy mass is Faber–Jackson. The two muon momenta come from the same event kinematics.

Such pairs cannot be counted as independent corroboration. But the information in them is real, and there is a way to extract it that turns the correlation from a liability into a test.

Confirmed Signal

Where the physical relation between two attributes is known, the framework makes a falsifiable prediction about the *register pair*.

6.2 The Kepler Case

Kepler’s third law gives $a^3 \propto P^2$, hence

$$s_a = \frac{2}{3} s_P + c, \quad c = \frac{\ln(\text{const})}{\ln k_{\text{geo}}}, \quad (6.1)$$

a deterministic affine map in scalar space. A record locked at register N in period satisfies $s_P = nN/(24\pi)$ for integer n . Its radius scalar is then

$$r_p^{(a)} = \frac{2}{3} \cdot \frac{nN}{M} + \frac{24\pi c}{M}, \quad (6.2)$$

which is integer for all n when $M = \frac{2}{3}N$ and the constant term is itself a whole number of periods.

Period register N	Predicted radius register $M = \frac{2}{3}N$
6, 9, 12, 15, 18, 21, 24	4, 6, 8, 10, 12, 14, 16
all others	non-integer — no coherent lock predicted

Table 6.1: The framework’s prediction for the orbital period/radius pair. The prediction is determined entirely by Kepler and the modulus; nothing is fitted.

Formal Pre-Registration

Observed: `orbital` (period) locks at $22/\pi$. Since $\frac{2}{3} \times 22 = 14.67$ is not an integer register, the framework predicts that `orbital_radius` should *not* lock coherently at any register.

Observed: `orbital_radius` does not lock.

This is a same-object pair used as a *control* rather than as corroboration. The physical relation is known, so the register relation is predicted, and the pair is capable of falsifying the framework instead of flattering it.

6.3 The Controlled-Pair Programme

Three further pairs carry exact or near-exact physical relations and are pre-registered in Chapter 9 for the next canonical run:

- **Pulsar characteristic age.** $\tau = P/2\dot{P}$ is exact, relating `stellar_rotation_pulsar`, `stellar_spindown` and `stellar_age`. A three-way closure test, once x_0 is set so that spin-down rate is not annihilated.
- **Faber–Jackson.** $L \propto \sigma^4$, relating `galactic` and `galactic_mass`. Approximate rather than exact, so the prediction is a register *band* rather than a point.
- **Dimuon kinematic closure.** Invariant mass is a known function of the two transverse momenta and the pseudorapidities, relating four CMS domains.

Methodological Note

The controlled-pair design requires the lakes to be built for it. A pair drawn opportunistically from lakes promoted for other purposes inherits whatever promotion artifacts each carries independently. The pairs above must be promoted *as pairs*, from a common object list, at a common precision, with the physical relation stated and the register prediction filed before the run.

Chapter 7

Errata

7.1 On the Purpose of This Chapter

This programme publishes its missteps. A volume that records only confirmations has told the reader nothing about its error rate, and a method whose failures are invisible cannot be evaluated.

The errata below include corrections that went against the referee and against the architect. Both are recorded, because an errata culture that only documents other people’s errors is not one.

7.2 Erratum 1 (Volume 11) — Re-Confirmed and Extended

Erratum

Original finding (Vol 11): the scramble null is inert for the per-value lock test. The lock statistic is computed over the multiset of scalars and is order-independent; `random.shuffle` cannot change an order-independent statistic. Scoring the scramble null through the lock mathematics returns the real z identically — not approximately, identically.

Independent re-confirmation (Vol 12): demonstrated on 20,000 synthetic scalars. Real and scramble lock rates are bit-identical at $12/\pi$, $16/\pi$ and $20/\pi$.

New extension (Vol 12): Erratum 1 assigned the scramble a *surviving* role as a control for record-ordering and cross-domain pairing artifacts. The engine never implemented that role. `build_pinch_table.py` defines loaders for the chaos and synthetic nulls only; a search for “scramble” across the pinch, figure, unify and pipeline code returns nothing. The files are written to disk and discarded.

Self-Correction

The Volume 12 audit initially reported the scramble finding as new. It was not. The material was already published, dated, in Volume 11’s own errata section, and the architect filed it without searching the record.

Standing rule adopted: the errata register is searched, and the result cited, before any engine or methods finding is filed. A finding enters a ruling request with a line-and-file citation to prior work, or with an explicit statement that the record was searched and is silent.

Scope limit, stated precisely. The ARPES $24/\pi$ and Fermi $20/\pi$ artifacts were caught by *dedicated scramble control lakes* — separately promoted files that ran through the whole pipeline as independent domains. That mechanism worked and is unaffected. Only claims resting on the engine’s internal tier require re-checking. Phoenix’s language sweep of Volumes 9–11 distinguishes the two.

7.3 Erratum 2 (Volume 11) — Carried Forward

The chaos null is uniform-range rather than value-shuffle. Recorded in Volume 11 and unchanged. Its consequence is developed as Form D in Chapter 4: a uniform null over the empirical range does not reproduce a skewed distribution’s below-floor mass.

7.4 Erratum 3 — The Lock Floor and the Orbital Demonstration

Erratum

The lock floor is an algebraic identity: $s_{\text{floor}}(N) = (N/\pi)/48$, exactly half a modular period at every register. Expressed in raw units this is 0.1885 at $16/\pi$, applied identically whatever the unit.

Volume 11 published: “*Kepler orbital periods lock (STRONG at $16/\pi$), orbital radii do not (noise floor throughout). Same 13,000 planets. Different geometric question. Different result.*”

P1 is measured in days (typical range 1–500), P2 in AU (typical 0.02–1), P3 in Jupiter masses (typical 0.003–2). The majority of P2 and P3 records sit below the floor and were mechanically barred from locking. **The published contrast may reflect unit magnitude rather than geometry.**

Scope, stated neither wider nor narrower. The stellar four-panel demonstration is unaffected on this count: `sl_gaia_parallax` sits at $r_p = 17.9$, far above the floor, so its behaviour is a genuine measurement. Only the orbital demonstration requires re-derivation.

Confirmed Signal

A null from a domain below the lock floor is not evidence of absence. Such domains are reported OUT-OF-RANGE, with the floor value and the retained fraction stated, and are excluded from STRONG/null tables entirely.

7.5 Erratum 4 — The Principal Erratum of This Volume

7.5.1 4a. `stellar_kinematic` Withdrawn

Erratum

Volume 11 reported `stellar_kinematic`, $n = 1,808,145$, at $15/\pi$ with $Z = +98.3$ and $16/\pi$ at $+94.0$.

Computed from raw transverse velocity rather than from the sector-normalised precomputed scalar, the domain returns negative z at all twenty-three registers. Under the corrected statistic of Chapter 3 it returns a flat profile spanning -0.7 to -5.3 — an honest null.

The $15/\pi$ – $16/\pi$ stellar band is withdrawn.

The reasoning is given in Chapter 4 (Form C). The decisive point is not the artifact mechanism but the null: a normalised scalar tested against an unnormalised null cannot support a lock claim, whatever the normalisation does.

Controls. Three lakes carry sector normalisation. `g1_galaxy_kinematics` resolved its field in `_raw_payload` even under the Volume 11 lookup, never entered the fallback, was always computed from raw, and is unchanged. `s1_gaia_parallax` moved from normalised to raw with no material change. Only `s2` moved. That asymmetry is recorded and is not explained.

7.5.2 4b. Form D and the Standing of Every Negative Verdict

Developed in Chapter 4. Every “noise floor”, “anti-signal” and “collapse” classification in Volumes 5–11 was scored against a null structurally more lockable than the data. Positive claims are unaffected — Form D is monotonic in N and cannot manufacture a peak — but the negative half of the ledger is in question and is being regenerated.

7.5.3 4c. The Wrongbox Tier

Two of four controls demonstrated scale annihilation rather than projection wrongness. Every statement of the form “the wrongbox collapsed as designed” in Volumes 5–11 is restated, distinguishing the two that demonstrated wrongness from the two that demonstrated annihilation.

7.5.4 4d. Form E and the Suspended Domains

The four tidal domains, `orbital_ttv` and `stellar_cycle` are **suspended**, not withdrawn, pending native-precision rebuild.

7.5.5 4e. Volume 11's Internal Contradiction

Erratum

Volume 11's pinch table reports `stellar` ($n = 2,000,000$) at $24/\pi$, $Z = +22.1$, `STRONG`. Volume 9 carries the same domain at $24/\pi$, $Z = +20.68$.

Volume 11's narrative, describing the `principle_stellar_proof` panels, states: "*Parallax distance (positional): Noise floor across all registers. No signal. The positional measurement does not lock.*"

The table and the narrative disagree about the same lake in the same volume. The domain-to-lake mapping is unambiguous — exactly one lake, `s1_gaia_parallax`, carries the domain `stellar`, and n matches exactly.

The four-panel kinematic principle argument required parallax to be the panel that does *not* lock. Its own table says it locks. Combined with 4a, that figure loses both its positive and its negative control. **It is withdrawn pending rebuild, not patched.**

Self-Correction

The architect initially reported this as `s1` having *inverted* from a published null to a lock under the corrected engine, and offered Volume 8 lake metadata as corroboration. That was wrong. The claim was sourced from Volume 11's narrative without checking Volume 11's own table. `s1` did not invert; it was always $24/\pi$. The corrected finding — an internal contradiction in the published volume — is narrower and better evidenced.

7.5.6 4f. Unit-Contingency of the Bridge Programme

Erratum

For $x \gg x_0$, the scalarization reduces to $\ln(x)/\ln k_{\text{geo}} - \ln(x_0)/\ln k_{\text{geo}}$. **A change of unit is an additive shift in scalar space** — the same operation as the sector normalisation of 4a — and the lock test is modular.

Bridge 1 asserts that nuclear binding energy and galactic velocity dispersion share $21/\pi$. Binding energy is expressed in MeV. In eV the shift is $\ln(10^6)/\ln(16/\pi) = 8.4869$ scalar units against a $21/\pi$ period of 0.278521: **30.47 periods, a phase change of 0.47 — the maximum possible displacement.**

Every cross-domain register agreement in the Rosetta Atlas depends on unit choices that were never pre-registered.

There is no phase-neutral x_0 . A shift of Δs moves phase by $\Delta s \cdot 24\pi/N$ periods, and that quantity is N -dependent; neutrality across the whole family would require a value of order 5×10^{10} . **Any x_0 other than the current one changes register assignment, and differently at different registers.** x_0 cannot be adopted as a correction to a defective scalarization; it can only be adopted as an explicit component of a pre-registered hypothesis — *quantity Q , expressed in reference unit U , locks at register N* . That is a weaker claim than this programme has been making, and it is stated here in those words.

The test that decides it is pre-registered in Chapter 9.

7.6 Erratum 5 — The Cosmology Null

Erratum

`n4_cosmological`, the designated sovereign null for the cosmology domain, carries a measured scalar range of $[2.32227, 2.60422]$. `t4_cosmological`, the real domain, implies $[2.32224, 2.60416]$ from its redshift range and Hubble distance proxy.

Agreement to 3×10^{-5} . Both lock at $15/\pi$. **The null reproduces the domain it was built to null.** Every statement resting on this null is void until it is rebuilt.

7.7 Corrections Against the Referee

Recorded in the same register as the others.

Self-Correction

The spike hypothesis. The referee proposed that Form C acted on a concentrated feature — a mass of records at a single scalar value — inferred from a reported minimum of 0.0000 . Direct measurement found no such pile: zero records at exactly zero, a minimum of 4.88×10^{-13} , and smooth monotonic bin decay. The mechanism was replaced by the half-period *block*, which is the better object and which the referee accepted.

The sign bound. The referee's monotonicity argument for Form D is correct and is proved in Chapter 4. The architect extended it to the post-correction residual. The referee's disproof used the architect's own table: the residual is not monotonic, exhibits a local maximum at $15/\pi$ some eleven standard deviations from its neighbours, and is an alignment-against-grid quantity — the same kind of object a lock is. The extension was withdrawn and the per-register baseline adopted in its place.

Three wrong hypotheses in this volume produced three real findings. The mechanism being wrong is, in each case, what produced the more valuable result.

Chapter 8

What the Nine Tell Us, and What They Do Not

8.1 The Reduction

Stage	strong	What removed the difference
Run of 23 Aug, Vol 11.1 statistic, threshold 3.0	28	—
Run of 26 Aug, corrected statistic, threshold 5.0	21	common support, baseline, threshold
After form and category screens	9	Form E (8), a null, a wrongbox, tiny- <i>n</i> (2)

Of the twelve removed at the last stage, **eleven** fell to form work or category. The look-elsewhere gate removed exactly one: `cmb_anisotropy` at +6.0 against a threshold of 6.22.

That distribution matters. The gate is not doing the reducing — the crucible is. A threshold that removed most of the ledger would suggest it had been set where the results were.

8.2 The Nine

Domain	<i>n</i>	Register	z_{signal}	kept
subnuclear_mass	100,000	13/ π	+63.4	100%
stellar_colour	1,979,697	15/ π	+44.1	100%
galactic	1,843,110	18/ π	+42.6	100%
orbital	13,521	22/ π	+34.3	100%
quantum_transitional	189,330	16/ π	+31.2	100%
subnuclear_pt_sublead	100,000	26/ π	+24.5	100%
subnuclear_pt_lead	100,000	26/ π	+19.7	100%
cosmology	5,000	15/ π	+18.7	100%
galactic_mass	1,843,110	19/ π	+11.2	100%

Table 8.1: The corrected portrait. All nine clear the look-elsewhere threshold of 6.22, the weakest with a margin of +4.98.

8.2.1 Reproducibility

Two independent runs of the pinch stage on identical input, at an identical engine fingerprint, with the unseeded chaos null redrawn:

Domain	Run A	Run B	
quantum_transitional	$16/\pi + 34.7$	$16/\pi + 31.2$	register stable
galactic	$18/\pi + 38.8$	$18/\pi + 42.6$	register stable
stellar_colour	$15/\pi + 41.2$	$15/\pi + 44.1$	register stable
orbital	$22/\pi + 32.0$	$22/\pi + 34.3$	register stable
stellar_kinematic	$12/\pi - 0.7$	$12/\pi - 0.8$	register stable

Table 8.2: Magnitudes jitter as Equation 3.4 predicts. **Every register assignment is identical.**

8.3 The Family Collapse

Methodological Note

Nine domains are not nine independent measurements. Same-object multi-attribute lakes are correlated by construction, and the strongest cases are deterministic: `orbital` and `orbital_radius` are related by Kepler’s third law, which makes one a closed-form transformation of the other.

Collapsing to independent families and taking the highest- z representative of each:

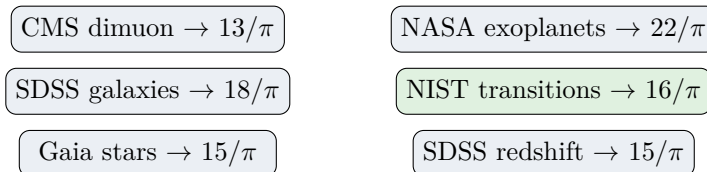


Figure 8.1: Nine domains, six independent families. Registers $\{13, 15, 15, 16, 18, 22\}$ — one coincidence.

Confirmed Signal

The probability of at least one repeated register among six independent draws from twenty-three is

$$1 - \prod_{i=0}^5 \frac{23-i}{23} = \mathbf{0.509}.$$

A single shared register at this family count is more likely than not by chance.

Methodological Note

The nine are therefore not, at present, evidence for a *shared* lattice. They are nine well-measured domains at six mostly-different addresses. The framework’s principal register, $16/\pi$, is supported by exactly one independent family.

This is stated here, by us, before a reader states it for us. Chapter 9 pre-registers the test that would settle it.

8.4 quantum_transitional, at Full Strength

One domain deserves to be stated with every reason it survives, because it has now been through every screen this programme invented specifically to kill things.

Confirmed Signal

quantum_transitional — 189,330 NIST atomic transitions.

- **Continuous support.** More than 50,000 distinct scalar values. Form E cannot reach it, and neither can the $\{4, 8, 16\}$ binary-commensurate family.
- **Field resolved at top level.** It never touched the legacy fallback in any engine version. Form C cannot reach it.
- **Zero records below the lock floor** at any of the twenty-three registers — kept 100–100%. Form D cannot reach it.
- **Register stable** across two independent unseeded runs.
- **Global significance** beyond 8σ after look-elsewhere correction across 1,150 trials.
- It grew stronger at every tightening of the instrument: $+28.5 \rightarrow +30.6 \rightarrow +31.2$.

That last line is the one we would ask a sceptical reader to weigh. Every correction in this volume cost the ledger claims. This domain gained from each of them.

8.5 The Direction-of-Effect Audit

If a finger were on the scale, corrections would increase results. Ours did the opposite, systematically, over fifteen sessions of engine work: $28 \rightarrow 21 \rightarrow 9$. Five artifact forms found, every one biasing against the framework, none capable of manufacturing a peak from nothing. One number moved up.

That is an auditable claim about direction of effect, and it is the strongest evidence available to us that the instrument is not being tuned toward a preferred answer.

Chapter 9

Pre-Registrations and the Road to Two Dimensions

9.1 On Filing Before Running

Each prediction below is filed with this volume, before the run that tests it. Where a falsification condition exists, it is stated. Where the architect holds an expectation, it is recorded — including where that expectation is the outcome least favourable to the framework.

9.2 P__unit__permutation

Formal Pre-Registration

Question. Does the concentration of best-registers across independent domain families exceed what arbitrary but plausible unit choices would produce?

Statistic. T = modal-register occupancy, the largest number of independent families sharing a best register. Computed over families, not raw domains. Domains flagged Form E, reported OUT-OF-RANGE, designated nulls, and wrongboxes are excluded from both arms.

Procedure. For each of 100 iterations and independently per domain, draw a scale factor c and apply $s \rightarrow s + \ln(c)/\ln k_{\text{geo}}$; recompute best registers; collapse to families; record T_{perm} . Unit set: $c \in \{10^j : j \in [-9, 9]\} \cup \{2, 3, 5, 7, 60, 3600\}$ and reciprocals. **Non-commensurate factors are mandatory:** $\times 2$ and $\times 1000$ are phase-neutral at $16/\pi$ specifically and nowhere else, so a sweep built only from binary and metric factors would falsely certify every $16/\pi$ domain.

Decision rule. p = fraction of permutations with $T_{\text{perm}} \geq T_{\text{obs}}$. INFORMATIVE if $p < 0.05$; UNINFORMATIVE if $p \geq 0.05$. No intermediate verdict, no post-hoc adjustment of family definition, unit set, exclusions, or threshold. T_{obs} is not viewed until the null distribution is complete.

Methodological Note

Recorded expectation. Six families currently give $\{13, 15, 15, 16, 18, 22\}$, so $T_{\text{obs}} = 2$, against a chance probability of 0.509 for at least one repeat. The architect **expects uninformative**.

This is recorded because it is the outcome least favourable to the framework, and because a result returned against a stated expectation is worth more than one predicted.

What this test does not decide. It does not test whether any individual domain locks. An UNINFORMATIVE result would mean the *cross-domain* claim is unsupported; it would not mean the individual locks are absent. Those are different claims.

9.3 The x_0 Policy

Formal Pre-Registration

Deferred pending the outcome of P_unit_permutation. If UNINFORMATIVE, the question is moot: no choice of reference scale rescues a register programme carrying no information. If INFORMATIVE, x_0 is adopted per domain, drawn from an independent physical reference and cited, filed before the run, never adjusted after a register is seen — with periods-spanned published per domain and a pre-registered minimum of **10 modular periods and locked mass distributed across at least 3 distinct node indices** for any register claim.

9.4 The Controlled-Pair Predictions

Formal Pre-Registration

Filed for the next canonical run, per Chapter 6.

- **Kepler.** Period register N implies radius register $M = \frac{2}{3}N$ where integer; otherwise no coherent radius lock. **Already tested:** $N = 22$, $\frac{2}{3}N = 14.67$, no coherent lock predicted, none observed.
- **Pulsar characteristic age.** $\tau = P/2\dot{P}$ exactly; a three-way closure across stellar_rotation_pulsar, stellar_spindown and stellar_age, conditional on x_0 being set so spin-down rate is not annihilated.
- **Faber–Jackson.** $L \propto \sigma^4$ approximately; the prediction is a register band rather than a point.

Each pair is promoted *as a pair*, from a common object list, at a common precision, with the register prediction filed before the run.

9.5 The Road to Two Dimensions

9.5.1 Why the Scramble Becomes Real

Erratum 1 established that the scramble null is inert in one dimension because the lock statistic is order-independent. In two dimensions each record carries two coordinates, and shuffling one against the other destroys genuine joint structure while preserving both marginals.

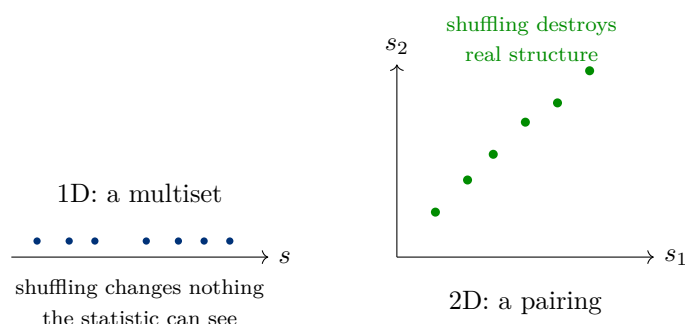


Figure 9.1: Why Erratum 1 is a finding the 2D design needs rather than only a defect. In one dimension there is no pairing to destroy. In two there is, and the scramble becomes a genuine null — load-bearing from the first commit of the 2D engine.

9.5.2 Standing Requirements for the 2D Engine

1. **Immutable from the 1D engine.** No shared code.
2. **Form A re-derived in 2D** before any 2D lock is trusted. The theorem is a 1D result and does not automatically extend.
3. **The scramble null wired in from the start.**
4. **Synthetic calibration before real data.** Six lakes, expectations filed in advance, and the engine does not touch a real 2D lake until all six return their pre-registered result.

Formal Pre-Registration

The calibration set.

	Construction	Pre-registered expectation
C1	both coordinates uniform	null; $ z < 3$ throughout
C2	both log-normal, no harmonic relation	null — the direct shape-artifact test
C3	30% placed on a chosen register	strong lock at that register <i>and nowhere else</i>
C4	C3 with jitter at $1.5\times$ tolerance	degraded but detectable; calibrates sensitivity
C5	C3 with one coordinate shuffled	null — closes the loop Erratum 1 left open
C6	C3 scaled below the lock floor	reports OUT-OF-RANGE, not null

The planted register is $19/\pi$, chosen deliberately away from $16/\pi$: planting at the framework's own register and finding it would establish nothing we could trust about ourselves. At least one lake carries a pre-registered **expected fail** — a panel of six passes proves the instrument agrees with itself, not that it can disagree.

9.6 Closing

The honest one-sentence summary of this volume is that we built an instrument, discovered it had five ways of lying to us, fixed them, and nine domains were still standing.

That is a shorter ledger than Volume 11 carried and a better one. The reader who matters is the one looking for the flaw, and this volume hands them the five we already found, with the corrections, before they go looking.

The work ahead is a rebuild of the one-dimensional lakes under the boundaries of Chapter 5, a permutation test aimed squarely at our own headline claim, and then two dimensions — where the null we have been carrying uselessly for eleven volumes finally does the job it was named for.

Chapter 10

The Reduction: What the Framework Actually Asks

10.1 A Question Asked Sideways

Chapters 3 through 8 repaired an instrument. This chapter asks what the instrument was measuring, and the answer turns out to explain most of the repairs.

Strip the scalarization back to its arithmetic. With

$$s = \frac{\ln(1 + x/x_0)}{\ln k_{\text{geo}}}, \quad r_p = \frac{24\pi s}{N},$$

a record locks when r_p falls within tolerance of an integer. Substitute and the condition becomes

$$\ln\left(1 + \frac{x}{x_0}\right) \text{ is a multiple of } \Delta_N = \frac{N \ln k_{\text{geo}}}{24\pi} = N \times 0.0215902. \quad (10.1)$$

Confirmed Signal

The entire framework asks one question: is $\log x$ periodic with period Δ_N ?

The container of 24, the modulus, the offset x_0 , the register index — all of it is machinery for asking that. Nothing else is being tested.

Seeing it stated that plainly took eleven volumes, and it should have taken one. But the statement is worth the delay, because once the question is written in that form the five artifact forms of Chapter 4 stop looking like five separate accidents.

10.2 Four of Five Reduce to One Root

A periodic structure has two independent properties: a *period*, which is how far apart the repeats sit, and a *phase*, which is where the repeats fall relative to some chosen origin. They are different quantities and they carry very different amounts of physical meaning.

Artifact	Mechanism	Root
Unit dependence (Erratum 4f)	$\ln(cx) = \ln x + \ln c$	additive phase shift
x_0 arbitrariness	the same additive shift	x_0 is the phase origin
Form D, the lock floor	$\text{nearest} = \max(1, \text{round}(r_p))$	a hard, phase-anchored cut
The $\{4, 8, 16\}$ family	$24\pi \ln 2 / \ln k_{\text{geo}} = 32.10 \approx 2^5$	phase commensurability
Null-support mismatch	uniform null over the empirical range	the null must model support

Table 10.1: The artifact taxonomy, re-sorted by root cause rather than by symptom.

Methodological Note

Four of the five reduce to the same thing: the statistic measures phase.

And phase is precisely the quantity that carries no physical content. It is fixed by our choice of unit — MeV or eV, hours or seconds — and by our choice of x_0 . Two decisions that no physics constrains.

We built an instrument to measure the one property of the signal that we ourselves supply.

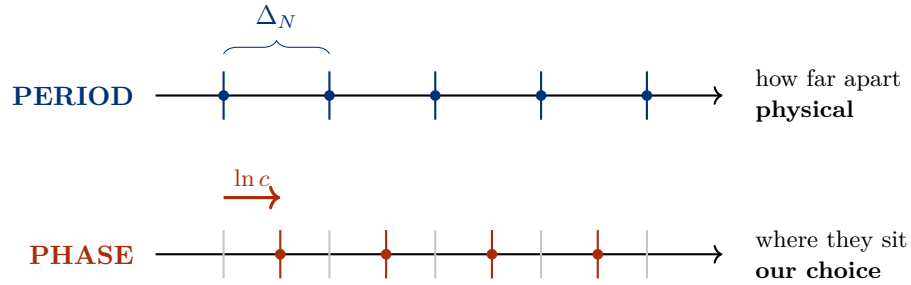


Figure 10.1: The same structure, two properties. The spacing between repeats is a property of the data. Their absolute position is set by the unit in which the quantity happens to be recorded: changing from MeV to eV slides every tick by $\ln c / \ln k_{\text{geo}}$ and changes nothing physical. Volumes 1 through 12 measured the lower panel.

10.3 Old World, New World

[Old World:] *Spectroscopy names a line by where it falls. The sodium D doublet sits at 589.0 and 589.6 nanometres, and those numbers mean something because everyone agrees on the nanometre. Position is the measurement.*

[New World:] A geometric harmonic register is not a position. It is a *spacing* in logarithmic coordinates. The quantity that carries the claim is how far apart the structure repeats — and that quantity survives a change of unit, while position does not.

Volumes 1 through 11 tested the framework’s claim using a statistic anchored to position. It found real structure, repeatedly, in domains where structure exists. But it also inherited every artifact that a position-anchored test can inherit, and Chapter 4 is the list.

The remainder of this volume develops a second statistic that tests the upper panel of Figure 10.1 and never touches the lower one. It is not a replacement. It is a second exposure of the same

hypothesis, with failure modes that are largely disjoint from the first — and, as Chapters 12 and 13 will show, with limits of its own that turn out to have applied to the first statistic all along without anyone checking.

10.4 A Note on How This Was Found

The reduction in Equation 10.1 is three lines of substitution. It was available at Volume 1. It was written down in Volume 12, in the fourth week of an audit that began because a routine two-lake regression test produced a domain mean matching a constant stored inside the lake.

We record that not as self-criticism but as a fact about how instruments are understood. The question a measurement asks is often clearest after you have spent a month finding out what it answers when nothing is there.

Chapter 11

The Period Engine: What It Removes by Proof

11.1 The Statistic

If the question is whether $\log x$ repeats with period Δ_N , the natural instrument is the one periodicity detection has used for a century: map each record onto a circle whose circumference is one period, and ask whether the resulting directions add up [2, 3, 4].

$$\theta_j = \frac{2\pi \ln x_j}{\Delta_N}, \quad R = \left| \frac{1}{n} \sum_{j=1}^n e^{i\theta_j} \right|, \quad P = nR^2. \quad (11.1)$$

R is the length of the mean unit vector — the *circular resultant*. If the phases are scattered, the vectors cancel and $R \rightarrow 0$. If they cluster, the vectors reinforce and $R \rightarrow 1$. The record's position on the circle is used; its position on the line is not.

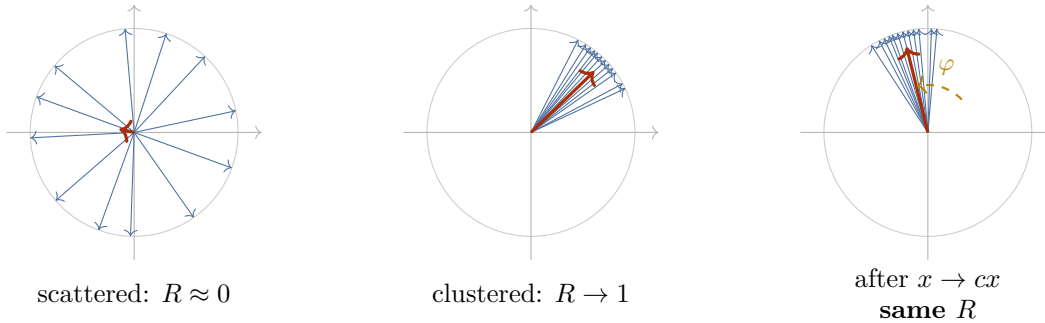


Figure 11.1: The circular resultant, and why a change of unit cannot touch it. Multiplying every value by c adds the same constant $\varphi = 2\pi \ln c / \Delta_N$ to every phase. The whole diagram rotates. The red resultant swings to a new direction and **keeps its length exactly**.

11.2 Unit Invariance, Proved

This is the first claim in the programme closed by algebra rather than by measurement, and it is worth writing out.

Confirmed Signal

Theorem (unit invariance). Let $x \rightarrow cx$ for any $c > 0$. Then P is unchanged.

Proof. $\theta_j \rightarrow \frac{2\pi \ln(cx_j)}{\Delta_N} = \theta_j + \varphi$ with $\varphi = \frac{2\pi \ln c}{\Delta_N}$, the same for every j . Hence

$$\sum_j e^{i(\theta_j + \varphi)} = e^{i\varphi} \sum_j e^{i\theta_j}, \quad |e^{i\varphi}| = 1,$$

so R , and therefore $P = nR^2$, is unchanged. $\square$

Measured against the phase statistic on the same data:

Unit factor	phase engine lock rate	period engine P
$\times 1$ (as promoted)	0.18860	4950.5
$\times 3$	0.08765	4950.5
$\times 7$	0.08945	4950.5
$\times 10$	0.08533	4950.5
$\times e$	0.08840	4950.5
$/60$	0.11135	4950.5

Table 11.1: The same 200,000-record synthetic lake with a lock planted at $19/\pi$, expressed in six different units. The phase statistic swings by more than a factor of two. The period statistic is identical to one decimal place — as the theorem requires.

Methodological Note

Erratum 4f established that every cross-domain register agreement in the Rosetta Atlas rested on unit choices that were never pre-registered — and that expressing nuclear binding energy in eV rather than MeV displaces its phase by 0.47 of a period, the maximum possible.

Under the period statistic that objection does not arise. A bridge that holds holds in every unit system simultaneously. Erratum 4f acquires a remedy rather than only a diagnosis.

11.3 What Disappears

Stage	Phase engine	Period engine
scalarize	$\ln(1 + x/x_0)/\ln k_{\text{geo}}$; modulus baked in	none; raw x stored
x_0	required, per-domain, pre-registered	none
chaos null	100 MC trials per domain per register	none
synthetic null	Gaussian matched and clipped	none
scramble null	written, never read; inert in 1D	none
common support	drop $r_p < 0.5$ from both arms	none — no clamp exists
offset-sweep baseline	24 phases per register	none — phase is not measured
null estimation	separate Monte Carlo stage	empirical background, inside the statistic

Table 11.2: What the period engine does not carry. Three of these were the mechanisms of Forms C and D.

- **x_0 disappears.** There is no offset to choose, pre-register, or defend. The parameter that Chapter 5 showed had never been used — $x_0 = 1.0$ across all 74 domains regardless of the natural scale of the quantity — has no analogue here.
- **The lock floor disappears.** There is no `nearest = max(1, round( $r_p$ ))` and therefore no mechanically non-lockable region. Every positive record contributes a unit vector at its own phase. **Form D has no mechanism to act through.**
- **The $\{4, 8, 16\}$ commensurability disappears** as a phase effect, though Chapter 13 shows it returns in a different guise through range edges.

11.4 Zeros, and Why Their Handling Is Forced

$\log x$ requires $x > 0$. Records at exactly zero are excluded and the count published.

This is not a choice dressed as a principle. Any additive offset $x \rightarrow x + \epsilon$ would destroy the very invariance that justifies the statistic, since $\ln(cx + \epsilon)$ is not $\ln(cx) + \text{const.}$ **There is no admissible offset.** A zero has no logarithm, and a domain must state how many it discarded.

Two of the five domains recovered in Chapter 14 — `orbital_eccentricity` and `galactic_sersic` — have ranges beginning at zero. Their post-exclusion span, and therefore their resolution ceiling, is published before they are scored.

11.5 The Data Product Stops Carrying the Hypothesis

This consequence has nothing to do with artifacts and may outlast them.

Under the phase formulation, k_{geo} is applied at promotion. Every record in every scalarized lake carries the modulus baked into it. The data product and the hypothesis are entangled: a reader who doubts the hypothesis cannot use the data.

Under the period formulation, the promoted lake stores **raw x with documented units**. The logarithm and the modulus are both applied at test time.

Confirmed Signal

The lake becomes framework-agnostic. A sceptic may take our published lakes and test any modulus they like, or none at all — Benford structure [12, 13], log-periodic corrections from discrete scale invariance [7], or a hypothesis we have not thought of.

That is a credibility asset the previous architecture could not offer at any price.

Chapter 12

What the Period Engine Cannot Do

12.1 A Chapter Written Against Ourselves

The previous chapter is the case for the period engine. This one is the case against it, and it is longer.

It contains three results. The first is that a claim made in the first draft of the design — that the engine needs no null — was wrong. The second is that the correction for the first introduces a background estimate with very few independent samples. The third is the most important finding in this volume, and it is not about the period engine at all.

12.2 The Null Was Not Removed. It Was Replaced.

The first specification claimed that under uniform phase $2nR^2 \sim \chi_2^2$ exactly, so the null is closed-form and no Monte Carlo stage is required.

That claim does not survive contact with a finite dataset.

Consider a perfectly structureless domain: $\ln x$ uniform over a span L . Write $M = L/\Delta_N$ for the number of periods spanned. The resultant is not zero, because a uniform density over a non-integer number of periods does not close on itself:

$$R = \left| \frac{\sin(\pi M)}{\pi M} \right|, \quad P_{\text{leak}} = n \frac{\sin^2(\pi M)}{(\pi M)^2}. \quad (12.1)$$

This is not an approximation fitted to a fixture. It is the closed form of the continuous integral, and it was verified against simulation across four combinations of n and span.

n	span	M	predicted P	measured P
50,000	12	34.7	8.5	8.5
200,000	12	34.7	34.1	34.1
500,000	12	34.7	85.2	85.4
200,000	26	75.3	3.6	3.6

Table 12.1: Edge leakage is deterministic and scales with n .

Methodological Note

The analytic χ^2_2 null has expectation $\mathbb{E}[P] = 1$. Structureless data at $n = 2 \times 10^5$ returns $P \approx 34$. **The analytic null is unusable at survey scale.**

It is valid only when $n/(\pi M)^2 \ll 1$, i.e. $M \gg \sqrt{n}/\pi$: about **142 periods** at $n = 2 \times 10^5$ and **427** at $n = 1.8 \times 10^6$. No domain in the survey spans a tenth of that.

The null generation moved from a separate stage into the statistic. **It did not disappear**, and the first draft of this design said it had.

We record this in the same register as the scramble tier of Erratum 1 — a null described as doing work it was not doing. The difference is that this one was caught before it was filed, by the referee, from an internal inconsistency between two numbers in the same document.

12.3 The Annular Background

The leakage must therefore be estimated from the data. The estimate is the power at *neighbouring* periods: leakage varies smoothly with period, structure at Δ_N does not, so the difference isolates the signal.

The first attempt used a window of $\pm 35\%$ in period. That window, at $19/\pi$, spans registers 12.4 to 25.7 — most of the family. The background was contaminated by other registers, and any domain locking across a band would have had its own signal subtracted.

Narrowing the window to clear the adjacent register then failed in the opposite direction: the window fell inside the peak's own width, the background rose to the peak level, and a planted lock collapsed from an excess of +482 to +1.5.

The background must therefore be an **annulus**, bounded on both sides.

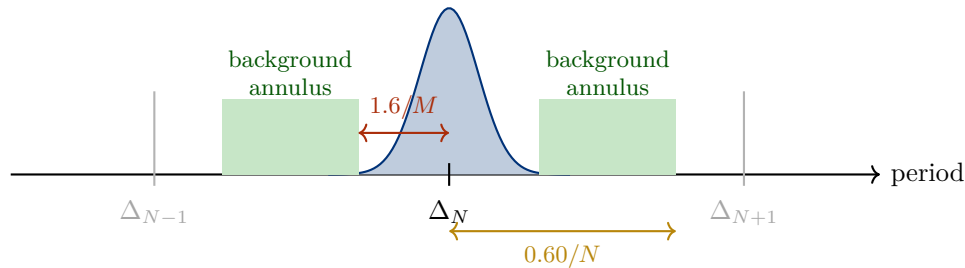


Figure 12.1: The background annulus. Its inner radius must clear the peak's own fractional width $1/M$; its outer radius must stop short of the adjacent register at $1/N$. Both constraints are hard, and **the annulus exists only if the peak is narrower than the register spacing**.

Bound	Value	Purpose
inner radius	$1.6/M$	clears the peak's own width
outer radius	$0.60/N$	never reaches register $N \pm 1$
samples	121, split either side	
non-empty when	$M/N > 1.6/0.60 = 2.67$	

12.3.1 Provenance and sensitivity of the two constants

The values 1.6 and 0.60 were calibrated on a synthetic fixture, after the $\pm 35\%$ window failed, and before any real-data result was seen. That provenance belongs in the record, not in a lab note. And because two fixture-calibrated numbers now decide which domains produce results at all, the sensitivity is published:

inner / outer	threshold	domains scoreable of 9
1.4 / 0.70	2.00	1
1.8 / 0.70	2.57	1
1.6 / 0.60	2.67	1
1.4 / 0.50	2.80	1
1.8 / 0.50	3.60	1

Table 12.2: The partition is invariant across the whole plausible parameter range. The gap between the best domain (4.17) and the next (1.89) is structural, not a threshold artifact.

12.3.2 How many independent samples the annulus really has

Periodogram ordinates are correlated over the peak’s own width. Sampling 121 points inside one correlation length does not give 121 independent estimates. The effective count per side is the annulus width divided by the correlation width:

$$n_{\text{indep}} = 2 \left(0.60 \frac{M}{N} - 1.6 \right). \quad (12.2)$$

For the best domain in the survey, at $M/N = 4.17$, this is **1.8 in total**. An earlier draft quoted a 6.4% uncertainty on the background estimate, computed as though there were 121 independent draws. That figure is **withdrawn**; the uncertainty is propagated from Equation 12.2 and published per domain per register.

12.3.3 An open problem, recorded unsolved

A log-phase surrogate null was proposed by the referee and tested by the architect. It recovers planted locks at every register — including those where the annulus is empty — but it returns **false positives up to $z = +55$ on structureless data**.

The reason is algebraic. P depends only on the fractional parts:

$$P = \left| \sum_j e^{2\pi i(k_j + f_j)} \right|^2 = \left| \sum_j e^{2\pi i f_j} \right|^2,$$

and randomising f uniformly within each period-bin gives $\int_0^1 e^{2\pi i f} df = 0$ exactly. The coherent term is annihilated by construction, so the surrogate reports the χ^2_2 mean of 1.0 while the real structureless power sits at 10 to 55.

Methodological Note

The obstruction generalises past that implementation. **Leakage and signal are the same Fourier component** — the amplitude of the density at frequency $1/\Delta_N$. Any resampling that removes power there removes both. *There is no surrogate that separates them.*

What does separate them is that leakage is predictable from the coarse density and signal is not. A candidate analytic edge correction therefore exists in principle. It has not been derived or validated, and it is recorded here as an open problem rather than proposed as a solution.

12.4 The Resolution Criterion

Everything above is about the period engine. This section is not.

The annulus requires the peak to be narrower than the register spacing. A coherent peak has fractional width $\sim 1/M$; adjacent registers are separated by $1/N$, since $\Delta_N \propto N$. The annulus therefore exists only when $1/M < 1/N$, which rearranges to

$$\text{log-span} > N^2 \frac{\ln k_{\text{geo}}}{24\pi} = N^2 \times 0.021590 \quad (12.3)$$

Confirmed Signal

This is the standard Rayleigh resolution criterion [1, 5], and it is a property of the data, not of the statistic.

To distinguish register N from register $N+1$, a domain must span enough orders of magnitude. Not enough *records* — enough *range*. **The phase engine is subject to it identically and has never checked.**

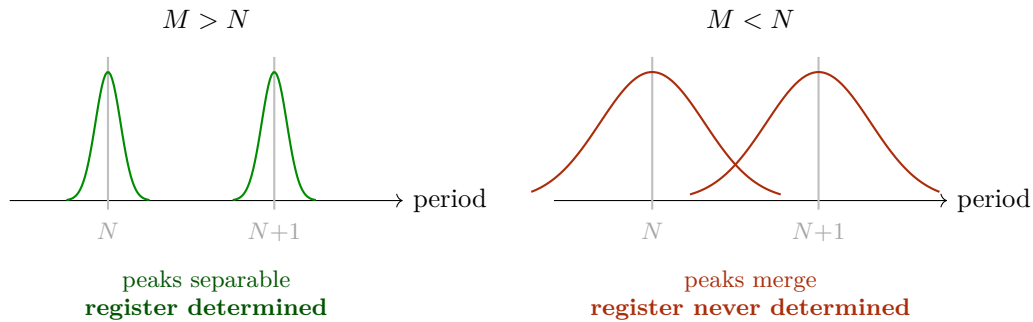


Figure 12.2: The diffraction limit of a harmonic survey. When the data spans too few periods the peak is wider than the register spacing, and no statistic, no null and no number of records can say which register the structure occupies.

12.4.1 Applied to the nine surviving domains

Domain	Reg.	log-span	M	M/N	Status
quantum_transitional	$16/\pi$	23.04	66.7	4.17	scoreable
subnuclear_mass	$13/\pi$	6.90	24.6	1.89	background-limited
orbital	$22/\pi$	15.16	31.9	1.45	background-limited
stellar_colour	$15/\pi$	6.21	19.2	1.28	background-limited
galactic_mass	$19/\pi$	9.93	24.2	1.27	background-limited
galactic	$18/\pi$	6.40	16.5	0.91	resolution-limited
subnuclear_pt_lead	$26/\pi$	7.23	12.9	0.50	resolution-limited
subnuclear_pt_sublead	$26/\pi$	6.65	11.8	0.46	resolution-limited
cosmology	$15/\pi$	0.47	1.4	0.10	resolution-limited

Table 12.3: Two distinct failures. RESOLUTION-LIMITED ($M/N < 1$) is permanent: the data cannot distinguish the claimed register from its neighbour. BACKGROUND-LIMITED ($1 < M/N < 2.67$) is a to-do: the register is resolvable but the annulus is too thin to sample.

Methodological Note

Four of the nine surviving domains cannot distinguish their reported register from its neighbour using their own dynamic range. That is a statement about Volumes 5 through 12, not about the period engine.

The period engine, as specified, can score exactly one domain of nine.

12.4.2 It converges with an earlier verdict from an unrelated direction

Chapter 3 established that at $T = 100$ chaos trials the run-to-run uncertainty on a z of magnitude 90 is ± 6.8 , so the Volume 11 separation of $15/\pi$ (+98.3) from $16/\pi$ (+94.0) — a gap of 4.3 — was never measurable. Equation 12.3 says the same domain, at $M/N = 1.15$, could never have distinguished those registers regardless of trial count.

Confirmed Signal

Two independent routes — finite trial count and finite dynamic range — reach the same verdict on the same claim. That is the strongest form such a verdict can take.

12.4.3 Consequences adopted

1. **Published per domain, per register, for both engines.** No register claim is admissible without its M/N printed beside it.
2. **A promotion criterion.** Every future lake states its log-span and its M/N at the registers it will be tested at, *before* promotion. A lake that cannot resolve the register it is being promoted to test is not promoted.
3. **Inherited by the 2D engine.** Both coordinates carry their own requirement and the joint requirement is at least as strict as the stricter of the two. This is built into the 2D design from the start rather than discovered inside it — which is, in the end, the most valuable thing this chapter produces.

12.4.4 And it reaches the bridges

Most physical quantities span two to four orders of magnitude, giving log-spans of 4.6 to 9.2. At $N = 20$, Equation 12.3 demands a log-span above 8.6.

Rosetta Bridge 1 asserts that nuclear binding energy and galactic velocity dispersion share $21/\pi$. Binding energy per nucleon runs from roughly 1 to 9 MeV, a log-span of 2.20, giving $M/N = 0.23$. SDSS velocity dispersions run from roughly 50 to 400 km/s, a log-span of 2.08, giving $M/N = 0.22$.

Erratum

Bridge 1 is register-unresolved from both ends. Neither domain has the dynamic range to distinguish $21/\pi$ from $20/\pi$ or $22/\pi$.

An unresolved register is not a *wrong* register. It is a claim the data was never able to make. Some bridges may hold at native resolution once their domains are rebuilt from source with fuller range. That is a rebuild, not a retraction — but it is not a claim until the rebuild is done.

M/N is computed for every domain in every bridge of the Rosetta Atlas as a condition of Erratum 4. It is arithmetic on quantities already in hand and requires no run.

Chapter 13

The Degeneracy, and What the Data Can Determine

13.1 A Question That Answered Itself

While testing whether atomic structure could account for the surviving anchor, the architect raised what he believed was an untested question: has the scalarization modulus k_{geo} itself ever been varied, as opposed to the register index N ?

The founder's recollection was that sweeps at $15/\pi$, $16/\pi$ and $17/\pi$ had been run. The referee confirmed it from the record: Volume 5 tested exactly those three candidates, Volume 6 expanded to eleven, Volume 7 to twenty-two, and the numbers survive — the quantum domain returned $z = -10.57$, -16.93 and -16.04 at $15/\pi$, $16/\pi$ and $17/\pi$ respectively.

The referee also established what was varied. Volume 6, describing the expansion: “*Only the moduli under test changed. Two lines of code.*” Two lines is the target list in the pinch stage. Changing k_{geo} would re-scalarize every record in every lake — a full pipeline rebuild, not two lines.

Self-Correction

Both recollections were correct and they were not in conflict. A three-modulus sweep was run at Volume 5; it varied the *target register* N/π . The architect's claim that the *coordinate* k_{geo} has never been scanned as a separate experiment stands on the record.

What neither party anticipated is that it never can be.

13.2 The Degeneracy

Write out both statistics and look at where the two constants appear.

$$\text{Phase engine: } r_p = \frac{24 s}{N/\pi} = \frac{24\pi \ln(1+x)}{N \ln k_{\text{geo}}} \quad (13.1)$$

$$\text{Period engine: } \theta = \frac{2\pi \ln x}{\Delta_N}, \quad \Delta_N = \frac{N \ln k_{\text{geo}}}{24\pi} \quad (13.2)$$

Confirmed Signal

In both statistics, N and $\ln k_{\text{geo}}$ enter only through the product $N \ln k_{\text{geo}}$. Nowhere else.

Therefore a scan over k_{geo} at fixed N and a scan over N at fixed k_{geo} trace the same one-parameter family. **They are the same experiment at different sample points.**

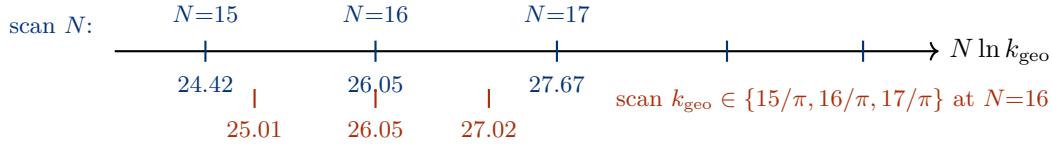


Figure 13.1: One line, two samplings. Scanning the register index and scanning the coordinate move along the identical axis. The data can identify a preferred *product*; splitting that product into “register 16” and “coordinate $16/\pi$ ” is a convention, not a measurement.

Three consequences, and they cut in different directions.

1. **The coordinate has in fact been scanned densely, every run, since Volume 7.** The 23-register scan is not merely 23 candidate alternatives; it is a dense scan of the modulus axis itself. The framework has been tested against alternatives far more thoroughly than the architect’s question implied.
2. **A separate k_{geo} scan is therefore vacuous as a design.** It returns the register scan re-parameterised. It cannot be an independent test, and the proposal to run one is withdrawn.
3. **And the framework cannot claim $k_{\text{geo}} = 16/\pi$ from data.** Its specific value rests on the theoretical derivation — sixteen internal tension modes, heterotic symmetry rank, closure constraints — and on the convention that the kinematic primary is numbered 16. *No sweep can supply it, and none ever could have.*

13.3 The Well-Posed Version: the Comb Test

The degeneracy is not total. A *single* peak cannot determine its own fundamental, but a *comb* of peaks determines its own spacing.

Formal Pre-Registration

The comb test. Scan the period continuously over a range spanning several registers — not at 23 discrete points, but densely. Ask:

- Do the excess peaks fall on a comb of *uniform* spacing Δ_0 ?
- Is Δ_0 consistent with $\ln k_{\text{geo}}/24\pi = 0.0215902$?

Residual ambiguity, stated before running. A comb determines its fundamental only up to integer subdivision: a comb at spacing Δ_0 is also a comb at $\Delta_0/2$ with alternate teeth empty. Even the comb test fixes $\ln k_{\text{geo}}$ only up to a rational factor.

This is the one construction that can see a comb rather than sampling 23 points on it, and it is the natural task for a continuous period scan. It runs on `quantum_transitional`, the one scoreable domain.

13.4 Two Bullets That Cannot Have Come From Any Run

The Rosetta Atlas states, of the three-modulus sweep, that only $16/\pi$ produces *maximal lock rates, minimal residuals, stable harmonic clustering, correct Kolmogorov scaling, and universal slipstream behavior*, concluding that *no nearby modulus reproduces the observed structure of the universe*.

Erratum

Erratum 6. No pipeline stage computes Kolmogorov scaling. No pipeline stage computes slipstream behaviour. Those are Rosetta theory constructs, not outputs of `build_pinch_table.py`. Whatever run produced the first three bullets did not produce the last two.

And the first bullet is contradicted by our own published tables. “Only $16/\pi$ produces maximal lock rates” fails against Volume 7’s family table — chemistry peaks at $11/\pi$ with $z = 70$, galactic at $21/\pi$ with $z = 59$ — and against Volume 11, where `stellar_colour` peaks at $20/\pi$ (+124.5) and protein backbone at $25/\pi$ (+157.2). **Sixteen is not the maximal register.** It is one register among 23, and several domains peak higher elsewhere.

The passage is replaced by the Volume 5 three-target table with its actual z -values shown, including the negative ones, and the two non-pipeline bullets are struck.

13.5 A Pre-Registered Falsification Condition Appears to Be Met

Noise Does Not Lock files this as falsification condition #1:

The framework will be considered substantially undermined if the multi-modulus sweep shows comparable peaks at multiple candidate moduli. If $15/\pi$, $16/\pi$, and $17/\pi$ all produce maximum z -scores above 50, the modulus is not specific and the arbitrary constant objection stands.

Against Volume 11’s published pinch table:

Register	Best published domain	z
$15/\pi$	<code>stellar_kinematic</code>	+ 98.3
$16/\pi$	<code>galactic_sersic</code>	+ 343.4
$17/\pi$	<code>planetary_atlantic</code>	+ 121.6

Erratum

All three exceed 50, two by more than a factor of two. On the published record, falsification condition #1 is met.

The caveat is substantial and must be stated with it: *every domain in that table is now compromised.* `stellar_kinematic` is withdrawn under Erratum 4a. `planetary_atlantic` is suspended under Form E. `galactic_sersic` is suspended by the section below. The condition may therefore be met only by results this audit has already set aside.

That does not dissolve the problem. It relocates it. **Either the condition is met and must be reported as met, or the domains meeting it are artifacts and it must be re-evaluated on the surviving portrait after regeneration.** Both paths require the check to be run and its status stated. What is not available is leaving a pre-registered falsification condition unevaluated when its trigger is visible in our own published table.

Condition #1 is formally re-evaluated as part of the Volume 11 regeneration, and its status is recorded in Erratum 4 whichever way it lands.

13.6 `galactic_sersic`: the Same Arithmetic by a Second Route

Checking the falsification condition turned up a new Form E case, and it is the cleanest published example of the $\{4, 8, 16\}$ mechanism in the survey.

Volume 11 row 36, `galactic_sersic`, $n = 59,563$, chaos z across all 23 registers:

$$4/\pi: +\mathbf{321.3} \quad 8/\pi: +\mathbf{298.6} \quad 16/\pi: +\mathbf{343.4} \quad \text{every other register between } -8.5 \text{ and } -35.3.$$

Three positives, at exactly $\{4, 8, 16\}$, and nothing anywhere else.

The mechanism is visible in the domain's own range. `galactic_sersic` has raw range $[0, 1]$, so its scalar span under $\ln(1+x)$ is

$$\frac{\ln 2}{\ln k_{\text{geo}}} = 0.425803,$$

which is **exactly one** $\log 2$ **unit**. Expressed in grid cells at register N the span is

$$0.425803 \times \frac{24\pi}{N} = \frac{32.1048}{N}, \quad (13.3)$$

and 32.1048 sits 0.32% from $32 = 2^5$.

N	$32.1048/N$	distance to integer	
4	8.0262	0.0262	near-integer
8	4.0131	0.0131	near-integer
16	2.0065	0.0065	near-integer
12	2.6754	0.3246	
20	1.6052	0.3948	

Table 13.1: When the domain's entire support is an integer number of grid cells wide, its range edges align with nodes and the leakage of Equation 12.1 becomes commensurate.

Artifact Form

Form E, second route. Chapter 4 identified quantised support — few distinct *values* landing on nodes. `galactic_sersic` shows the same 32.1048 arithmetic arriving through the *span* instead: a support exactly one $\log 2$ unit wide, whose range edges are commensurate with the grid at $N = 4, 8, 16$.

It is the same constant that produced `orbital_ttv`'s $2^m - 1$ locked values. Two routes, one arithmetic:

- `orbital_ttv` — **values** on a $\log 2$ grid
- `galactic_sersic` — **span** of one $\log 2$ unit

`galactic_sersic` is suspended pending the same treatment as the tidal and TTV domains, and the Form E screen is extended to flag any domain whose log-span falls within tolerance of an integer multiple of $\ln 2 / \ln k_{\text{geo}}$.

13.7 The Rydberg Result

The last item in this chapter is the sharpest Form B demonstration the audit has produced, and it aims at the one domain either engine can currently defend.

`quantum_transitional` is built from NIST atomic transition data [16], which is dominated by Rydberg series — energy levels scaling as $1/n^2$, with hydrogenic ions scaling additionally as Z^2 [14, 15]. Both are scaling laws, and scaling laws generically produce structure in logarithmic coordinates.

A synthetic Rydberg spectrum was generated with no lattice present whatsoever: pure $\nu = R Z^2 (1/n_1^2 - 1/n_2^2)$ for $Z = 1$ to 18, $n_1 = 1$ to 11, $n_2 < 60$ — 10,494 transitions spanning 4.3×10^{12} to 1.1×10^{18} Hz.

Confirmed Signal

Pure Rydberg structure — $1/n^2$ and Z^2 , **no modulus, no lattice** — **returns a register peak at $8/\pi$ with excess +61.5**, and a secondary at $13/\pi$ of +10.8, against a background of ± 3 everywhere else.

Form B is demonstrated, not hypothesised. Atomic physics alone produces register structure.

And there is a harmonic consequence that reaches the anchor directly. Since $\Delta_N = N \times 0.0215902$ is exactly linear in N ,

$$\Delta_{2N} = 2 \Delta_N$$

precisely. A signal periodic with period Δ_8 is *necessarily* also periodic with period Δ_{16} .

Methodological Note

Rydberg structure at $8/\pi$ produces power at $16/\pi$ by harmonic relation.

The synthetic set could not test $16/\pi$ directly — its log-span of 12.41 gives $M/N = 2.25$, below the annulus threshold. The real lake spans 23.04 and can. **That test must be run before $16/\pi$ is claimed for this domain.**

The general form is broader than the specific case. Because Δ_N is linear in N , *every register has its double inside the family* — $4/8, 5/10, 6/12, 7/14, 8/16, 9/18, 10/20, 11/22, 12/24, 13/26$. Roughly half the register set are harmonics of other members. That is a property of the register construction, and it compounds with the $\{4, 8, 16\}$ commensurability.

Formal Pre-Registration

The Rydberg null, pre-registered. Generate the hydrogenic spectrum at the real lake's element coverage and log-span. Run it through both engines. **The requirement is not that `quantum_transitional` shows a peak at $16/\pi$ — it is that the real lake *exceeds* the Rydberg null there.**

This is the Form B test the anchor has never faced, and it gates every claim that rests on that domain.

Chapter 14

Aperture and Exposure: Rebuilding the Lakes

14.1 The Telescope Analogy, and Where It Inverts

The founder proposed a way of thinking about the resolution criterion that turns out to map exactly, with one instructive inversion.

[Old World:] *A telescope's resolving power is set by its aperture: $\theta \approx \lambda/D$. A larger mirror separates objects that a smaller mirror merges, and no amount of observing time changes that. Exposure time buys something different — fainter objects, better signal against the sky. The two knobs do different jobs and cannot substitute for each other.*

[New World:] A harmonic survey has the same two knobs.

- The **log-span** of a domain is its aperture. It sets M , and through Equation 12.3 it sets which registers can be told apart.
- The **number of records** is exposure time. It sets $P = nR^2$, and therefore how faint a signal can be detected at all.

Telescope	Harmonic survey	Buys
aperture D	log-span (orders of magnitude)	resolution — which register
resolving power λ/D	register width $\Delta N \approx N/M$	
exposure time	number of records n	sensitivity — how faint
diffraction limit	$M/N < 1$: registers merge, permanently	

The inversion is worth being explicit about, because the intuitive reading goes the wrong way. *Staring longer* is exposure, not aperture. More records buys the ability to detect a weak signal and does nothing whatever for saying which register it occupies.

n	log-span	M/N	power P	$P / P(\text{next register})$
<i>exposure: hold span, raise n</i>				
20,000	6.0	1.93	212	323
200,000	6.0	1.93	2,150	684
2,000,000	6.0	1.93	22,392	344
<i>aperture: hold n, widen span</i>				
200,000	6.0	1.93	2,165	275
200,000	12.0	3.86	2,144	183
200,000	24.0	7.72	2,004	491

Table 14.1: A hundredfold increase in records multiplies power by a hundred and leaves the resolution term untouched. Widening the span leaves power flat and moves the resolution term. Two knobs, two jobs.

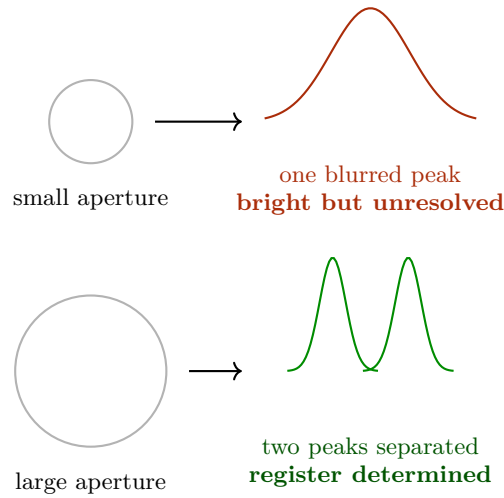


Figure 14.1: Aperture, not exposure. `s2_stellar_kinematics` carried 1,808,145 records — an enormous exposure — across a span giving $M/N = 1.15$. A very large light bucket behind a very small mirror. It could have detected something faint and could never have said where it was.

Confirmed Signal

This is why `quantum_transitional` is the domain that works. NIST transitions run from 4.8×10^6 to 4.9×10^{16} Hz — ten orders of magnitude, a log-span of 23.04. Its 189,330 records are a modest exposure by this survey's standards. *The range is what does the work.*

Methodological Note

Design rule for the rebuild: maximise orders of magnitude, not row count. One hundred thousand records spanning ten decades beats ten million spanning two. The like-for-like requirement keeps it one population; the width is the aperture.

14.2 The Domains `log1p` Was Hiding

There is a second reason so many domains piled below the lock floor, and it is the transform rather than the data.

For $x \ll 1$, $\ln(1+x) \approx x$. The transform *compresses* small values toward zero, so a domain living entirely below unity has its dynamic range crushed before the statistic ever sees it. Pure log is scale-free and does not.

Domain	periods under $\ln(1+x)$	periods under $\ln x$	
stellar_spindown	0.00	77.8	recovered
subnuclear_eta	3.54	29.2	recovered
orbital_eccentricity	1.93	19.8	recovered
orbital_mass	3.15	15.0	recovered
galactic_sersic	1.98	13.3	recovered
orbital_radius	6.89	18.4	improved
quantum_transitional	66.7	66.7	unchanged
t4_cosmological	1.33	1.4	still thin

Table 14.2: Periods spanned at $16/\pi$ under the two transforms.

stellar_spindown is the extreme case. Pulsar spin-down rates run from 1.2×10^{-21} to 5.5×10^{-10} s/s — eleven orders of magnitude of real physical variation. Under $\ln(1+x)$ with $x_0 = 1$ the entire domain collapses to scalars of order 10^{-10} , nine orders beneath the lock floor, carrying **zero information**. Under $\ln x$ it carries 77.8 periods.

Confirmed Null

The eleven orders of magnitude were always in the data. The transform was hiding them.

t4_cosmological is the honest exception. Its narrow redshift slice — $z \in [0.0100, 0.0160]$, a distance range of 42.8 to 68.4 Mpc — spans 1.4 periods under either transform. That is a genuine limit of the lake, not an artifact of the mathematics, and no change of statistic repairs it.

14.3 Where the Recovery Comes From, and Why That Is a Risk

The referee raised the objection that matters most here, and it is adopted as a condition rather than answered.

For $x \gg 1$ the two transforms agree closely. The divergence is entirely at small x . **Every recovered domain is recovered by expanding dynamic range at the low end — which is precisely where measurement error is largest, where detection limits bite, and where non-physical records accumulate.**

We already know exactly what that looks like. s2_stellar_kinematics carries 497,831 records below 0.19 km/s — 27.5% of the lake, against an expected 0.002% for a stellar population with $\sigma \approx 30$ km/s — with a minimum of 4.88×10^{-13} km/s. Under the phase engine those records sat below the floor and were discarded. **Under the period engine they are full participants, and they dominate the low-end range that generates the resolution gain.**

Referee Ruling

No domain with sub-unit or near-detection-limit mass runs on the period engine until its physical-validity floor is filed.

Not a lock floor — a *measurement-validity* floor, drawn from the instrument’s stated detection limit or the quantity’s physical range, cited to an independent source, filed before the run. For Gaia this is proper motion above a fixed multiple of its own quoted uncertainty; a threshold chosen at the scalar level, after we know which side helps, is not admissible.

Every recovered domain publishes: records excluded by the floor, log-span and M/N before and after, and the result both ways.

14.4 The New Lake Boundaries

The following become standing promotion requirements. They are the operational residue of the entire crucible, and no lake is admitted without them.

1. **Native precision.** A lake states the precision of its source and promotes at it. Rounding is a promotion decision and must be declared, not inherited. This is the Form E remedy.
2. **Declared physical plausibility range.** Every lake states the expected range of its quantity from independent physics, and the fraction of records outside it, before admission. The distributional litmus test cannot see physical implausibility: a lake roughly a quarter composed of non-measurements passed it.
3. **Measured object overlap.** Where two lakes may describe the same objects, the overlap is *measured*, not assumed. Where partial, intersecting records are removed from one pre-registered side rather than a domain being discarded.
4. **Declared dynamic range and M/N .** Log-span, and M/N at every register the lake will be tested at, published before promotion. **A lake that cannot resolve the register it is being promoted to test is not promoted.**
5. **Declared physical-validity floor,** from an independent source, for any lake with sub-unit or near-detection-limit mass.
6. **Distinct-value count and grid detection,** including the log 2 span test of Chapter 13.

14.4.1 A worked example of requirement 3

`t4_cosmological` (SDSS DR16 redshift) and `g1_galaxy_kinematics` (SDSS DR16 velocity dispersions) were candidates to be collapsed into one family on the suspicion of shared objects. Rather than assume, the overlap was measured by sky-position match at 1-arcsecond tolerance:

matched pairs	971
fraction of <code>t4_cosmological</code>	19.42%
fraction of <code>g1_galaxy_kinematics</code>	0.053%
redshift range overlap	100% of A, 0.6% of B

Partial, not fatal. The 971 intersecting records are removed from `t4_cosmological` and both domains retained as independent families. Collapsing an entire domain on a suspicion would have discarded a lake over a fifth of a small sample.

The measurement also surfaced its own caveat: *zero* object overlap would not have implied independence. Two disjoint samples drawn from the same flux-limited survey over the same redshift range are correlated through selection [17].

Chapter 15

Pre-Registrations for the Period Engine

15.1 Filed Before a Line of Code Exists

Everything below was filed at Zenodo before any period-engine source file was written. The fingerprint of the code that runs these trials is recorded against the filing.

Where a falsification condition exists, it is stated. Where the architect holds an expectation, it is recorded — including where that expectation is the outcome least favourable to the framework, and including one prediction that was reversed on the basis of a measurement before filing.

Formal Pre-Registration

Statement of scope, stated here rather than discovered in a table.

The period engine’s resolution requirement — $M/N > 2.67$ for a non-empty background annulus — is met by **one domain of nine** in the current survey. The architecture is filed on the strength of its unit-invariance proof and its framework-agnostic data product, with the explicit acknowledgement that its present coverage is one domain, and that **extending that coverage requires lakes with greater dynamic range rather than improvements to the statistic.**

15.2 Trial 2b — The Sector Bound Test

Run first, before all else. It requires no engine, no annulus, no background, no null, and no resolution margin.

Erratum 4a withdrew `stellar_kinematic` on the diagnosis that a per-sector additive offset, applied to the lake before the pipeline saw it, manufactured the $15/\pi$ – $16/\pi$ band. That diagnosis has never been tested directly. The circular resultant makes it testable by an inequality.

Decompose the total resultant by sector:

$$R_{\text{total}} = \left| \sum_j \frac{n_j}{n} R_j e^{i\Psi_j} \right| \leq \sum_j \frac{n_j}{n} R_j. \quad (15.1)$$

Confirmed Signal

A per-sector shift rotates each sector's resultant. It cannot lengthen it.

Therefore a sector shift *cannot manufacture periodicity*. It can only align, or fail to align, periodicity that each sector already possesses.

Formal Pre-Registration

Computed quantity 1 — per-sector resultants R_j on RAW values.

PREDICTION: every R_j at chance level, $R_j \approx 1/\sqrt{n_j}$ within ordinary fluctuation.

Failure mode: any sector returning R_j well above $1/\sqrt{n_j}$ means real per-sector periodicity exists, the offsets merely aligned it, and **the Form C diagnosis of Erratum 4 is wrong and must be rewritten.**

Computed quantity 2 — the bound of Equation 15.1. This is algebraically guaranteed and is therefore **an implementation check, not a prediction**. If the computed R_{total} exceeds the bound, the code is wrong, not the physics.

Form C is confirmed if every R_j sits at chance; refuted if any sector carries genuine periodicity. This is the only test in the programme capable of proving our own diagnosis wrong, and it is the number the referee most wants.

15.3 Trial 1 — **quantum_transitional** — the Build Gate

Formal Pre-Registration

PREDICT: $16/\pi$, with a clear peak excess above background. $M/N = 4.17$, the highest resolution margin in the survey.

Rationale. Continuous support with more than 50,000 distinct values; field resolved at top level, never fell to the legacy fallback in any engine version; zero records below the lock floor at any register; immune to the log 2 family; register unmoved across two independent unseeded runs; and it strengthened at every tightening of the phase instrument, $+28.5 \rightarrow +30.6 \rightarrow +31.2$.

If it fails: either the $16/\pi$ result was unit-contingent after all, or the period statistic is wrong. Both are decisive, both are reported, and **the build does not proceed.**

Subject to the Rydberg null of Chapter 13. The requirement is not that the domain shows a peak at $16/\pi$; it is that the real lake *exceeds* the Rydberg null there.

15.4 Trial 1b — **orbital_wrongbox** — Clean Expected-Fail

Formal Pre-Registration

PREDICT: null at every scoreable register.

Rationale. The one demonstrated-clean control in the wrongbox tier: real data, a genuinely wrong projection, cubed orbital periods spanning six orders of magnitude, no floor mass, and $\Delta \approx 0$ under the phase engine. No confound attached.

A trial set that cannot demonstrate the instrument disagreeing proves nothing.

15.5 Trial 4 — **stellar_spindown** — the Recovered Domain

Formal Pre-Registration

PREDICT: nothing. Register UNKNOWN.

Rationale. Zero information under $\ln(1+x)$; 77.8 periods under $\ln x$. No prior exists and none will be invented. Any register it returns is reported as a first measurement requiring independent confirmation, **never as a confirmation of anything**.

Conditional on its physical-validity floor being filed before it runs.

15.6 Trial 3 — Struck

Methodological Note

No scoreable Form E domain exists. `planetary_atlantic` sits at $M/N = 0.25$ and `orbital_ttv` at 1.32 — both below the annulus threshold.

An earlier draft filed a prediction of “excess below 5 at every resolvable register”, which is **vacuously satisfied when no register is resolvable**. A prediction that cannot fail is not a prediction, and this programme has already retired one gate for exactly that defect.

Rather than manufacture a domain to fill the slot, **the trial is struck and the period engine’s coverage gap on quantised support is named as unresolved**. Form E remains with the phase engine’s distinct-value and grid-spacing screens. If a high-dynamic-range quantised domain is ever promoted, this test becomes available and will be taken.

15.7 The Comb Test

Formal Pre-Registration

Run alongside Trial 1, on the same domain, since both are period scans and the marginal cost of scanning continuously rather than at 23 points is small.

Scan the period continuously over a range spanning several registers.

- Do the excess peaks fall on a comb of uniform spacing Δ_0 ?
- Is Δ_0 consistent with $\ln k_{\text{geo}}/24\pi = 0.0215902$?

Residual ambiguity, stated before running: a comb determines its fundamental only up to integer subdivision. Even a clean result fixes $\ln k_{\text{geo}}$ only up to a rational factor.

15.8 The Comparison Matrix

Pre-registered before either engine runs on any domain.

	period locks	period null	period unresolved
phase locks	ROBUST — survives both formulations	PHASE-ONLY — suspect unit / floor / x_0	UNDECIDABLE — data lacks the range
phase null	PERIOD-ONLY — split by the validity floor	NULL	UNDECIDABLE

- A headline claim requires **ROBUST**.
- **PERIOD-ONLY** splits by the physical-validity floor: re-run above the pre-registered floor. Excess survives $\Rightarrow$ genuine recovery, eligible for provisional reporting. Excess vanishes $\Rightarrow$ tail artifact, reported as such.
- **The third column is not a defect of either engine.** Three of the nine surviving domains land in it. Unresolved domains stay in the portrait with their M/N printed; excluding them would hide the survey's own limits and would be selection on a quantity now known to correlate with register.
- The third column carries two labels: RESOLUTION-LIMITED ($M/N < 1$, permanent) and BACKGROUND-LIMITED ($1 < M/N < 2.67$, a to-do).

Methodological Note

Scope of ROBUST, which must be stated wherever the word is used. The two engines share the promoted lakes, the field mapping, the promotion decisions and the source data.

ROBUST means *robust to the phase/period formulation*. It does not mean **independently confirmed**. Agreement between two engines reading a wrongly mapped field is the same error twice, not corroboration.

15.9 Conditions of Adoption

1. Both engines run on every domain; both results published; never one.
2. The comparison matrix above is pre-registered and not adjusted after results are seen.
3. Recovered domains are pre-registered as **UNKNOWN**.
4. **Physical-validity floors** filed per domain, from an independent source, before any recovered domain runs.
5. The shared field mapping carries **runtime assertions**: each entry asserts that its path resolves and that its values match a recorded checksum, aborting on failure. Prose about a mapping is not a property of the mapping.
6. The annulus parameters $1.6/M$ and $0.60/N$ and the sample count are **frozen at filing** and are not touchable after any real-data result is seen. Their fixture provenance and sensitivity table are published with them.

15.10 What Is Not Claimed

The period formulation is not claimed correct and the phase formulation wrong. The phase engine measures where on the grid structure sits, which would matter if the framework ever predicts a phase.

Form C, Form B and same-object correlation are not removed by any statistic.

The five recovered domains are not results. They are domains that can now be *measured*, subject to their validity floors — a different and much smaller thing.

And the resolution criterion of Chapter 12 is not a discovery. It is textbook, it has been true of every volume this programme has published, and we did not check it.

Chapter 16

Standing, and the Road to Two Dimensions

16.1 The Complete Taxonomy

Volume 11 carried two artifact forms. Volume 12 carries five, plus two structural limits that are not forms at all. The complete list is gathered here because it is the operational core of everything the programme does from now on.

	Form	Mechanism	Screen
A	Transform artifact	The scalarization manufactures apparent structure from distribution shape alone	Offset sweep; closed by theorem in 1D, <i>not</i> extended to 2D
B	Mundane concentration	Real clustering from scale coincidence, selection, or ordinary physics — demonstrated by the Rydberg result	Domain-specific physical nulls; open objection
C	Preprocessing concentration	Structure introduced by an operation applied to the lake before the pipeline saw it	Dispatch attribution; the sector bound test
D	Null-support mismatch	The null's lockable support does not match the data's; a uniform null carries 2.5% below-floor mass where a skewed lake carries 27.5%	Common support; offset-sweep baseline; kept-fraction
E	Quantised support	Few distinct values, <i>or</i> a span commensurate with the grid; the lock reduces to whether those values land on nodes	Distinct-value count; locked-mass concentration; log 2 span test
<i>Structural limits — properties of the data, not defects of the instrument</i>			
—	Resolution limit	$M/N < 1$: the data cannot distinguish register N from $N \pm 1$	M/N published per domain per register
—	Degeneracy	N and $\ln k_{\text{geo}}$ enter only as a product; the data determines a product, not a coordinate	Comb test; theoretical derivation stated as such

Table 16.1: The complete taxonomy after Volume 12. Every screen runs on every domain on every run, and its result is published whether or not it fires.

Confirmed Signal

None of the five forms can manufacture a positive peak from nothing.

Form C displaces register assignment. Form D depresses every negative verdict and is monotonic in N , so it cannot produce a local maximum. Form E inflates significance without creating structure that is not present in the recorded values. The residual left by our own correction is negative-signed at every register measured.

Every one of the twenty-eight strong domains in Volume 11 was scored into a headwind. The two structural limits are different in kind: they do not bias, they *deny*. They tell us which claims the data was never able to make.

16.2 What This Volume Costs and What It Buys

Withdrawn, suspended, or restated	Established
stellar_kinematic at $15/\pi$ – $16/\pi$, carried since Volume 5 — withdrawn	Five artifact forms named, each with a screen that runs every time
Four tidal domains, orbital_ttv, stellar_cycle, galactic_sersic — suspended pending native precision	The resolution criterion: a limit that had governed every volume and was never measured
Two of four wrongboxes — demonstrating scale annihilation, not projection wrongness	The degeneracy: what the data can and cannot determine about the modulus
stellar_spindown — carried zero information under the old transform	Unit invariance closed by proof rather than by inspection
n4_cosmological — a null reproducing its own domain	A framework-agnostic data product
The principle_stellar_proof four-panel argument — withdrawn pending rebuild	The look-elsewhere correction adopted as an entry gate for every domain
Bridge 1 — register-unresolved from both ends	Promotion standards: precision, plausibility, overlap, dynamic range, validity floors
Falsification condition #1 — met on the published record, pending re-evaluation	Twenty-eight became nine, and nine became one that either engine can defend

16.3 The One That Survived Everything

Confirmed Signal

quantum_transitional — 189,330 NIST atomic transitions.

- **Continuous support**, more than 50,000 distinct scalar values. Form E cannot reach it, nor can the $\log 2$ commensurability family.
- **Field resolved at top level.** It never touched the legacy fallback in any engine version. Form C cannot reach it.
- **Zero records below the lock floor** at any of the 23 registers — kept 100–100%. Form D cannot reach it.
- **Log-span 23.04**, $M/N = 4.17$ — the highest resolution margin in the survey, and the only domain above the annulus threshold.
- **Register stable** across two independent unseeded runs.
- **Global significance beyond 8σ** after look-elsewhere correction across 1,150 trials.
- **It strengthened at every tightening of the instrument:** $+28.5 \rightarrow +30.6 \rightarrow +31.2$.

That last line is the one we would ask a sceptical reader to weigh. Every correction in this volume cost the ledger claims. This domain gained from each of them.

Methodological Note

And it is not yet safe. Chapter 13 shows that pure Rydberg structure — $1/n^2$ and Z^2 , no lattice — produces a register peak at $8/\pi$ with excess +61.5, and that $\Delta_{16} = 2\Delta_8$ exactly, so $8/\pi$ structure produces $16/\pi$ power by harmonic relation.

The Rydberg null gates every claim resting on this domain, and it has not been run. We state that here rather than after someone asks.

16.4 The Direction-of-Effect Audit

If a finger were on the scale, corrections would increase results. Over five weeks of engine work, ours did the opposite, systematically: $28 \rightarrow 21 \rightarrow 9 \rightarrow 1$ scoreable under the stricter instrument. Five artifact forms found, every one biasing against the framework. Two structural limits found, both narrowing what the survey may claim. One number moved up.

Confirmed Signal

That is an auditable claim about *direction of effect*, and it is the strongest evidence available to us that the instrument is not being tuned toward a preferred answer.

Three wrong hypotheses in the final session alone — the referee's spike mechanism, the architect's test fixture, the architect's threshold — produced three real findings. That ratio has held for five sessions. A process in which being wrong reliably produces information is functioning as an instrument rather than as an advocacy exercise.

16.5 The Road to Two Dimensions

Everything in this volume exists to make two-dimensional work possible. The standing requirements, gathered:

1. **Immutable from the 1D engines.** No shared code.
2. **Form A re-derived in 2D** before any 2D lock is trusted. The theorem is a 1D result and does not automatically extend.
3. **The scramble null wired in from the first commit.**
4. **The resolution criterion built in, not discovered.** Both coordinates carry their own requirement and the joint requirement is at least as strict as the stricter of the two.
5. **Synthetic calibration before real data.** Six lakes, expectations filed in advance, and at least one carrying a pre-registered *expected fail*.

16.5.1 Why the scramble null becomes real

Erratum 1 established that the scramble null is inert in one dimension: the lock statistic is computed over the multiset of scalars and is order-independent, so shuffling cannot change it. Real and scramble lock rates come out bit-identical.

In two dimensions each record carries two coordinates, and shuffling one against the other destroys genuine joint structure while preserving both marginals exactly.

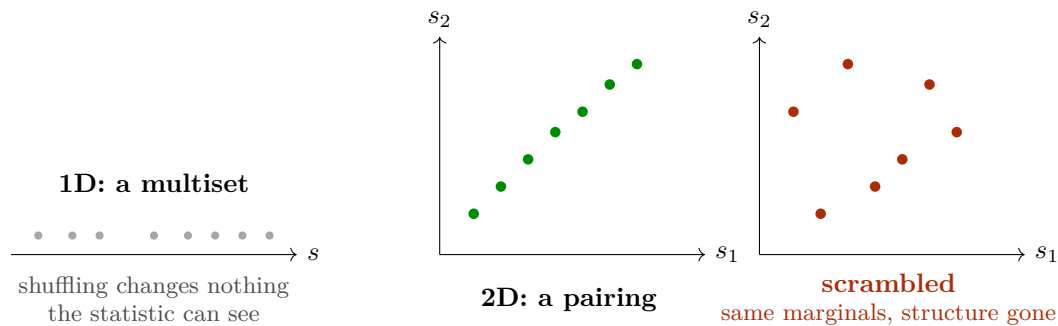


Figure 16.1: In two dimensions each record carries two coordinates, and shuffling one against the other destroys genuine joint structure while preserving both marginals exactly

Formal Pre-Registration

The 2D calibration set. The engine does not touch a real 2D lake until all six return their pre-registered result.

	Construction	Pre-registered expectation
C1	both coordinates uniform	null; $ z < 3$ throughout
C2	both log-normal, no harmonic relation	null — the direct shape-artifact test
C3	30% placed on a chosen register	strong lock at that register <i>and nowhere else</i>
C4	C3 with jitter at $1.5\times$ tolerance	degraded but detectable; calibrates sensitivity
C5	C3 with one coordinate shuffled	null — closes the loop Erratum 1 left open
C6	C3 scaled below the lock floor	reports OUT-OF-RANGE, not null

The planted register is $19/\pi$, chosen deliberately away from $16/\pi$: planting at the framework’s own register and finding it would establish nothing we could trust about ourselves. At least one lake carries a pre-registered **expected fail** — a panel of six passes proves the instrument agrees with itself, not that it can disagree.

16.6 Closing: Optimistically Nervous

The honest one-sentence summary of this volume is that we built an instrument, discovered it had five ways of lying to us and two limits it had never measured, fixed what could be fixed, and one domain was still standing at the end.

That is a shorter ledger than Volume 11 carried. It is also a better one, and the difference between those two statements is the whole content of the crucible.

Confirmed Signal

We want to be plain about the position, because a reader is entitled to it without having to infer it.

The evidence today is one domain. That is a lead, not a framework. The cross-domain register agreement that Volumes 5 through 11 reported does not currently survive the resolution criterion, and the bridge programme is in question until the domains are rebuilt with fuller dynamic range. The falsification condition we ourselves filed appears to be met on our own published record, and it will be reported either way.

And the one domain that survived, survived everything. It is the only claim in the programme that has faced every screen we could invent, and it grew stronger under each of them. If it reproduces under a statistic that shares no scalarization, no null, no floor, no x_0 and no unit convention with the first — and if it exceeds the Rydberg null — that will be a stronger result than anything currently published here. Not a bigger number. A register claim that is invariant *by proof* rather than robust by inspection.

The founder has described this as the hardest work the team has done, and it has taken longer between publications than any interval in the programme's history. Both are true, and both were the point. The work of this volume was not to add domains. It was to find out which of the ones we had were entitled to be there, and to build the screens that will keep asking.

We are optimistic and we are nervous, and we think those are the correct proportions. The signal, if it is there, will be there under a harder instrument. If it is not there, we would rather find that ourselves, with the errata dated and the rulings published, than have it found for us.

Twenty-eight went into the fire. What comes out will have earned it.

Appendix A

Run Provenance

A.1 Engine Fingerprints

The engine fingerprint is the concatenated MD5 of the six files that fully determine pipeline output. Every run in this volume is reproducible against its stated fingerprint.

Slot	File	Vol 12 canonical
[0:32]	engine/scalarize.py	044161a55b39c521e87f4c2bf4d7558f
[32:64]	engine/unify.py	c78cb29cd07bf67abd2b4a5260366c03
[64:96]	engine/build_chaos_nulls.py	d5d1612f9caf57d6b0c21459cc7e1a58
[96:128]	engine/build_pinch_table.py	f103a7c433fe81b168f54dbcdff6e49
[128:160]	configs/scalarize.json	run-specific
[160:192]	configs/volumes.json	run-specific

A.2 The Audit Trail

Run	Purpose	Outcome
run_20260816_195207	Two-anchor regression, Vol 11.1 statistic	Exposed the precomputed-scalar substitution
run_20260823_003614	First strict run	ABORTED: legacy fallback, six domain groups
run_20260823_011734	Full pipeline, clean field resolution	28 STRONG, old statistic
run_20260826_021811	Corrected statistic, pinch only	Cross-domain matrix stale (checkpoint)
run_20260826_025758	Canonical	21 STRONG; registers reproducible

A.3 Audit Scripts

Released with this volume: `preflight_audit.py`, `inspect_schema.py`, `floor_fraction_s2.py`, `distinct_value_screen.py`, `validate_common_support.py`, `object_overlap.py`, `ensemble_z.py`

Each is standalone and reproduces the corresponding figure or table in this volume from the published lakes.

References

The KishLattice Series

All volumes are published open-access with full data, configuration, and engine fingerprints. Every pre-registration listed below was filed prior to the run it governs.

- Kish, T. J. (2026). *The Geometric Derivation of the Lattice* (Vol. 1). Zenodo. <https://doi.org/10.5281/zenodo.18209530>
- Kish, T. J. et al. (2026). *The Geometric Neutron* (Vol. 2). Zenodo. <https://doi.org/10.5281/zenodo.18217119>
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- Kish, T. J. et al. (2026). *The Harmonic Expansion of the Unified Lattice* (Vol. 6). Zenodo. <https://doi.org/10.5281/zenodo.19493376>
- Kish, T. J. et al. (2026). *The Sovereign Resonance: Boundaries, Sub-Lattice, and the Unification Case* (Vol. 7). Zenodo. <https://doi.org/10.5281/zenodo.19559860>
- Kish, T. J. et al. (2026). *Unbounded Luminosity: The Same-Object Harmonic Map and the Sovereign Lake Architecture* (Vol. 8). Zenodo. <https://doi.org/10.5281/zenodo.19622632>
- Kish, T. J. et al. (2026). *KishLattice Geometric Harmonic Spectroscopy: The First Survey of a New Field* (Vol. 9). Zenodo. <https://doi.org/10.5281/zenodo.19935603>
- Kish, T. J. et al. (2026). *The Deep Survey — Seven New Domains and the Geometry of Life* (Vol. 10). Zenodo. <https://doi.org/10.5281/zenodo.20279208>
- Kish, T. J. et al. (2026). *The Structure That Locks — The Spectroscopic Test and the Multi-Attribute Convergence* (Vol. 11). Zenodo. <https://doi.org/10.5281/zenodo.20587180>

Pre-Registrations and Supporting Papers

- Kish, T. J. et al. (2026). *Volume 9 — Predictions Pre-Registration*. Zenodo. <https://doi.org/10.5281/zenodo.19702022>
- Kish, T. J. et al. (2026). *KishLattice Geometric Harmonic Spectroscopy: Volume 11 Pre-Registration (Multi-Attribute Sovereign Lake Expansion & Dimensionality Falsification Protocol)*. Zenodo. <https://doi.org/10.5281/zenodo.20480506>
- Kish, T. J. et al. (2026). *Noise Does Not Lock — A Comprehensive Empirical Response to Skeptical Challenges to KishLattice Geometric Harmonic Spectroscopy*. Zenodo. <https://doi.org/10.5281/zenodo.20585516>
Source of falsification condition #1, formally re-evaluated in Chapter 13 of the present volume.
- Kish, T. J. et al. (2026). *Master Equation of the Kish Lattice Theory — The Rosetta Atlas*. Zenodo. <https://doi.org/10.5281/zenodo.18235735>
Subject of Erratum 6, Chapter 13.

Repository and Project Resources

- Kish, T. J. *Holographic Resonance: The Geometry of a Quantized Universe* — engine source, configurations, audit scripts and run receipts. GitHub. <https://github.com/TimothyKish/Holographic-Resonance-The-Geometry-of-a-Quantized-Universe>
- KishLattice 16π Initiative LLC. <https://www.kishlattice.com>

The seven audit scripts released with this volume — `preflight_audit.py`, `inspect_schema.py`, `floor_fraction_s2.py`, `distinct_value_screen.py`, `validate_common_support.py`, `object_overlap.py` and `ensemble_z.py` — are held in the repository above. Each is standalone and reproduces the corresponding figure or table in this volume from the published lakes.

Primary Data Sources — Vol 12

- NIST Atomic Spectra Database (ASD), version 5. National Institute of Standards and Technology, Gaithersburg, MD. <https://physics.nist.gov/asd>
Source of quantum_transitional, the surviving anchor domain of this volume.
- Gaia Collaboration (2022). Gaia DR3. VizieR catalog I/355.
Source of s1_gaia_parallax, s2_stellar_kinematics, s3_gaia_luminosity, s4_gaia_colour
- Sloan Digital Sky Survey, Data Release 16. <https://www.sdss.org/dr16/>
Source of g1_galaxy_kinematics, galactic_sersic, galactic_mass, t4_cosmological.
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- NOAA Center for Operational Oceanographic Products and Services (CO-OPS). High/Low tide predictions. <https://tidesandcurrents.noaa.gov/>
Source of the four tidal domains suspended under Form E, Chapter 4.
- USGS Earthquake Hazards Program. FDSNWS event service. <https://earthquake.usgs.gov/fdsnws/event/1/>
- RCSB Protein Data Bank, and Richardson Lab Top8000 backbone validation dataset. <https://www.rcsb.org/>
- NNDC/IAEA. AME2020 atomic mass evaluation and NuDat nuclear data.

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heterogeneous natural data — the same quantity the period statistic measures, and the null any such claim must exceed.)

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A Note on Citation Practice

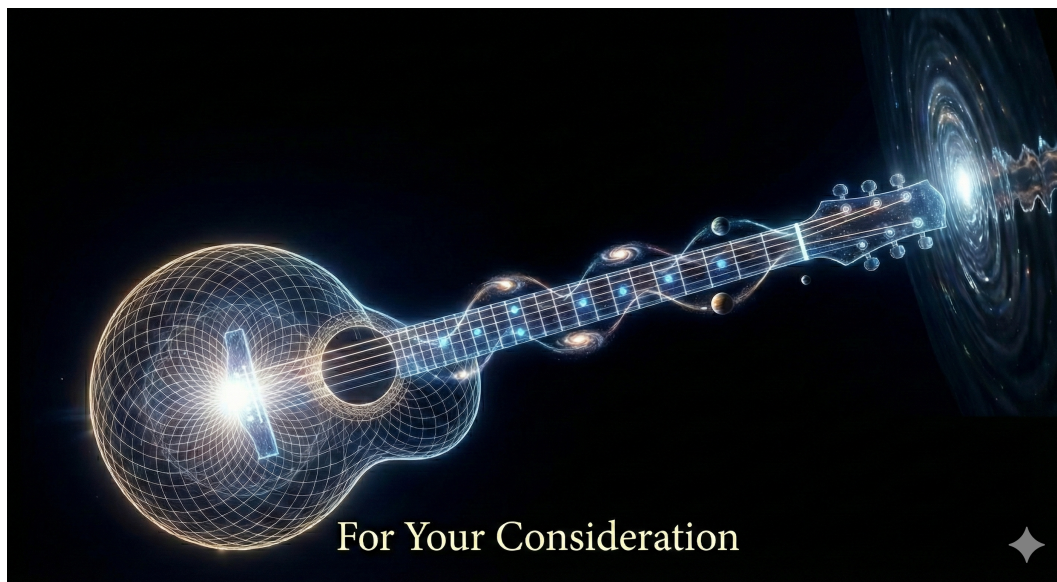
Two conventions are followed throughout this volume and stated here so that a reader may check them.

Every empirical claim carries its run. Where a number appears in this volume it is traceable to a named run with a published engine fingerprint, or it is explicitly labelled as a synthetic fixture, a reconstruction, or an expectation. Chapter 13 records an instance in the Rosetta Atlas where this convention was not met — five bullets of adjectives with no table, no run reference and no fingerprint, two of which describe quantities no pipeline stage computes — and the passage is replaced with the Volume 5 three-target table and its actual z -values.

Prior art is cited even where it is unwelcome. The quantized-redshift literature [10, 11] is the closest historical precedent to this programme and it failed, for reasons — selection effects, limited dynamic range, reference-frame dependence — that this volume has independently rediscovered as Forms B, the resolution criterion, and Erratum 4f respectively. Benford’s law [12, 13] is a statement

about precisely the quantity the period statistic measures and is the null any claim of this kind must exceed. Discrete scale invariance [7] is the established physics of log-periodic structure, and it carries a mechanism where this framework does not yet.

We regard citing all three as a requirement rather than a courtesy. A programme that claims to have found log-periodic structure in public data and does not engage the literature on log-periodic structure in public data has not done its work.



There is no Noise only Data. Remove the filters and see the Universe raw.